

16 Sole (*Solea solea*) in Subarea 27.4 (North Sea)

sol.27.4 – *Solea solea* in Subarea 27.4

16.1 General

The assessment of sole in Subarea 27.4 is performed using the Stock Synthesis 3 (SS3) platform (Methot and Wetzel, 2013), as agreed at the benchmark held in February 2024 (WKBFLATFISH; ICES, 2024b). During the benchmark, data on landings and discards, on weights-at-age, and indices of abundance from three surveys – BTS Q3, Sole Net Survey (SNS) and Demersal Young Fish Survey (DYFS) – were combined with updated values for natural-mortality and maturity at age. The assessment now explicitly includes age-0 fish and applies age-reading-error adjustments to all ages, based on error matrices computed during the 2016 exchange workshop (ICES, 2016). Model settings were later adjusted to improve the fit to the discards series and the indices of abundance in recent years (ICES, 2024a).

16.1.1 Stock structure and definition

North Sea sole is assumed to consist of a single stock unit, but indications exist of some level of spatial structure and separation. Stock identity of sole stocks was investigated in the second quarter of 2022 using genetic markers (SNPs) (Maes et al., 2025). Sole was collected on board commercial fishing vessels from 3 different rectangles in Division 27.4c (31F1 close to the Thames estuary; 31F2 and 31F3 close to the Belgian coast). Results showed significant genetic differentiation with samples collected in neighbouring areas (7d, 7e). Within the southern North Sea (4c), subtle significant population sub-structuring was detected between sole collected near the Thames estuary (31F1) and Belgian coast (31F2). No genetic differentiation was detected for sole from 4c and statistical rectangle 29F0 (7d), samples from 31F1 and 30E9 (7d), and 31F3 and 28E9 (7d). The combined result of limited larval connectivity and restricted adult dispersal, particularly during late spring/early summer (i.e., collection time of sole in this study), might explain some of the sub-structuring detected in the southern North Sea and English Channel.

16.1.2 Ecosystem aspects

North Sea sole is commonly distributed in the southern North Sea. Spawning takes place in spring in shallow waters along the coasts (Van der Land, 1991) following a latitudinal gradient (Rijnsdorp and Vingerhoed, 1994). Episodic large recruitment events take place at irregular intervals.

16.1.3 Fisheries

The Dutch fleet accounts for the majority of North Sea sole landings, mainly using beam trawls with mesh sizes of 70–99mm. From 2011 onwards many beam-trawl vessels switched to electrical pulse gear. In 2011, 42 pulse-trawl licenses were issued under two EU derogations, and a further 42 were added in 2014, bringing the total to 84. The catch composition of pulse trawls differs from traditional beam trawls (ICES, 2018). Pulse-gear adoption shifted effort towards softer grounds in the southern part of Division 4.c.

Following the European Parliament decision in 2019 to ban pulse fishing, only 5% of each member-state fleet could continue pulse fishing until 1 July 2021; a full ban has applied since then. Fleet responses have varied, influenced also by rising fuel prices. The number of vessels landing sole has declined from >200 in 2002 to ~90 in 2022, with widespread decommissioning of beam/pulse vessels since 2021 and continuing in 2023.

BMS landings of sole reported to ICES are currently much lower than the estimates of catch below the minimum conservation reference size (MCRS). In 2024, 17 t of BMS landings were reported (0.4% of catches) and 395 t of discards were estimated from observer and self-sampling programmes (9.1 % of catches).

16.1.4 Management regulations

Management of sole in subarea 4 is implemented through an annual TAC and technical measures.

The minimum landing size is 24 cm. Sole in the North Sea has been under the landing obligation since 2016 (EU 2018/973). Survivability and *de minimis* exemptions apply to this stock (EU, 2023/2459). The survivability exemption applies to common sole below MCRS caught with otter trawls (OTB) with a cod-end mesh size of 80–99 mm in ICES Division 4.c, within 6 nautical miles of the coast but outside identified nursery areas, by vessels with a maximum length of 10 meters and a maximum engine power of 221 kW, fishing in waters with a depth of 30 meters or less and with tow durations of no more than 90 minutes. The *de minimis* exemption is of use to common sole caught with (1) gillnets and trammel nets in Union waters of ICES division 3a, and ICES subarea 4, and should not exceed 3% of total annual catches of that species, and (2) beam trawls of mesh size 80–119 mm equipped with a Flemish panel in subarea 4.

The EU has from July 2021 banned the use of electric pulse fisheries in European waters (EU, 2019/1241). Pulse fishing had been adopted by a total of 84 Dutch vessels, which have since returned to beam trawling, switched to other gears or have been decommissioned.

ICES recognizes the EU multi-annual management plan (MAP) for this stock (EU, 2018), which ICES considers precautionary, although evaluated with an operating model assuming different productivity and reference points. There is no agreed shared management plan with UK for this stock, so advice is given according to the ICES MSY approach, with catch scenarios consistent with the MAP FMSY range also provided. Since 2021, ICES advice for this stock has been based on the ICES MSY approach rather than the MAP.

16.2 Fisheries data

16.2.1 Official landings

The official landings by country are presented in Figure 16.1. The landings have been fluctuating around 10 000 tonnes from 2006–2018, and have then declined sharply since 2019, with landings in 2024 being the lowest in the time-series. A time-series of the official landings by country and overall total, the officially reported BMS landings, and the agreed TACs is presented in Table 16.1. The TAC has not been overshoot since 2016 and has been substantially undershot in recent years (between 38 and 60% in the 2020–2023 period). This low uptake is atypical for a target species such as sole. TACs from 2021 were driven up by the estimated increase in spawning biomass brought by the strong but overestimated 2018 year class.

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In 2024, when a low TAC, based on a correction of abundances due to the strong retrospective pattern (ICES, 2024) was in place, official landings amounted to 102% of the TAC, while ICES-estimated total catch reached 118%. Official BMS landings in 2024 amount to 4.5% of the estimated discards.

16.2.2 ICES catch estimates

The entire catch time-series 2002–2022 was re-estimated during WKBFLATFISH (ICES, 2024b) to avoid some detected inconsistencies in the substitution and raising of samples across years. Raising and age-allocation procedures were coded in *R* scripts rather than using the InterCatch interface, and the same procedures have been applied to the 2024 data (Figures 16.2-4. The 2021–2023 values have also been recalculated following recent corrections on UK entries in the Inter-catch database for that period.

The Dutch fleet plays an important role for this stock, both in terms of landings and sampled strata. Considering the large impact of Dutch sampling, the Dutch discards self-sampling programme was evaluated (ICES 2024b). Every year, observer trips are carried out on boats taking part on the discards self-sampling programme, and their results then compared. For sole, the results indicate that self-sampling might be leading to underestimated amount of discards. A majority of the compared hauls showed larger discard estimates from observer data than from self-sampling, but haul and yearly averages showed a wide range, indicating the existence of error in the estimates rather than a constant bias. It was pointed out that the current self-sampling protocol is better tailored to other stocks, such as plaice, and could be improved for sole and other less common species.

Information on potential unreported catches and discard survival is currently not incorporated in the assessment.

Figure 16.5 presents the time series of total ICES catches, landings and discards over the 1957–2024 period. Discards were included in the SS3 assessment model as a bycatch fleet and were not reconstructed prior to 2002. Past reconstruction should be explored in the future. Total landings in 2024 were 3960 t, which is 2 % lower than in 2023.

Landings, in numbers-at-age (1–10+), as used as input for the assessment, are presented in Table 16.2 and Figure 16.6. Discards, in numbers-at-age (1–10+), as used as input for the assessment, are presented in Table 16.3 and Figure 16.7. The proportions of discarded sole by age over the 2002–2024 period, over which data on discards is available, is presented in Figure 16.8.

16.3 Weights-at-age

Weights-at-age in the landings of sole in Subarea 27.4 can be found in Table 16.4 and Figure 16.9. These are measured weights from the various national catch and market sampling programs. Discard weights-at-age (Table 16.5 and Figure 16.10) are derived from the various national catch and discard programs (both observer and self-sampling).

Stock weights-at-age are generated from the extracted quarter 2 weights at age in catch, and the same value is used by Stock Synthesis at the start and end of the spawning season (Table 16.6). Mean stock weights have shown a continuous downward trend, reaching (and lately dipping below) the levels seen at the start of the series (Figure 16.11). For the younger ages, weights declined but appear to be increasing again in recent years (Figure 16.12). This trend has been reversed in the last two years, with values in 2024 showing a large increase in mean weights-at-

age across most ages (Figure 16.13). The increase on stock weights-at-age is the main cause of the increase in estimated SSB in 2024 from the current assessment, when compared with last year's forecast.

16.4 Maturity and natural mortality

A knife-edged maturity-ogive with full maturation at age 3 is assumed for sole in Subarea 27.4 (Table 16.7). The benchmark (ICES 2024b) explored different data sources to update this ogive. Van Keeken et al. (2004) investigated maturity over the period 1965–2005 and showed that the proportion of mature females in quarter 1 at age 2 and 3 increased from 1995 onwards: for age 2 from 0% to 25–50% mature and for age 3 from 50% to almost 100% mature. However, more recent Belgian and Dutch maturity data from commercial catches could not provide updated estimates due to the low proportion of immature individuals in the catches. Additionally, assuming that age 2 fish are still immature which could result in a potential underestimation of the SSB, is considered more precautionary in light of the absence of better data. The benchmark recommended to collect more maturity data at age in the first quarter from a broad length range, for example from discards or surveys (with smaller mesh sizes) to modify the currently used ogive in the future. This maturity assumption results in large year-classes causing large jumps in SSB as they reach age 3.

Natural mortality follows the size-based Lorenzen (2022) formulation:

$$M_a = W_a^{-0.288}$$

where W_a is mean stock weight-at-age. Weights came from a von Bertalanffy curve ($L_\infty = 40.4$, $K = 0.33$, $t_0 = 0$) and the length–weight relation $W = 0.0068 L^{3.09}$. The whole vector is scaled to the mean $\bar{M} = 4.899 T_{\max}^{-0.916}$ (Then et al., 2014) using $T_{\max} = 26$ y (the oldest validated fish in BTS was 28 y). No time variation in M is included.

16.5 Survey data

Three survey series are used in the assessment of North Sea sole:

- Quarter 3 Beam Trawl Survey (BTS), covering the 1990–2024 period and containing samples for ages 0 to 10+.
- Quarter 3 Sole Net Survey (SNS), extending from 1990 to 2024, with the exception of 2003, and with samples including ages 0 to 6.
- Demersal Young Fish Survey (DYFS), extending from 1990 to 2024 and with samples including ages 0 and 1.

Data for all three has been extracted from the ICES DATRAS database.

Two indices of abundance are constructed from those surveys. The first one (BTS) uses the BTS Q3 samples collected by The Netherlands, Belgium, the United Kingdom and Germany (Figure 16.14). A standardized age-based index is calculated using a delta-lognormal GAM model, using the methodology presented in Berg *et al.* (2014). The second index (COAST) uses SNS (sampled by The Netherlands) and DYFS (sampled by The Netherlands, Belgium and Germany), and is calculated using a delta-lognormal GAM model, standardized for ages 0–5. Please refer to the WKBFLATFISH report (ICES, 2024b) for further details on the analysis.

A standardized comparison of the two indices over the available time-series is presented on Figure 16.15, while Figures 16.16 and 16.18 and Tables 16.8a and 16.9a present each individual index in their actual scales at age. The internal consistency plots of the cohorts of the two indices are

presented in Figures 16.17 and 16.19. Changes over the last three decades on the internal consistency of the coastal survey index are shown in Figure 16.20. The actual values of the two survey indices used in the assessment are presented in Tables 16.8b and 16.9b. Following a recent update that incorporated a number of past samples from the Belgium BTS to DATRAS, the full dataset was updated. The addition of these data has a large effect on the scale of the indices of abundance for the 2004-2024 period, and the reason for this is not clear at the moment. Further work is being conducted to fully understand the effect of those samples in the GAM standardization, so they are currently not part of the dataset used to build the BTS index of abundance.

In addition to the age based abundance, the estimated total biomass of the two delta-lognormal GAM indices are also used in the assessment, functioning as a scaling parameters over the total magnitude. The retrospective analysis of the numbers at age of BTS and COAST are illustrated in Figure 16.21 and 16.22 respectively.

16.6 Age reading error

Since the 2024 benchmark, bias in age reading was incorporated in the assessment model. A time-invariant age reading error matrix, obtained from the exchange exercise in 2016 (ICES, 2016), was applied to age data for the entire time series. The age reading error matrix is presented below.

Modal age	0	1	2	3	4	5	6	7	8	9	10
0	0.972	0.028	0	0	0	0	0	0	0	0	0
1	0.006	0.988	0.006	0	0	0	0	0	0	0	0
2	0	0.007	0.986	0.007	0	0	0	0	0	0	0
3	0	0	0.021	0.941	0.038	0	0	0	0	0	0
4	0	0	0	0.201	0.774	0.025	0	0	0	0	0
5	0	0	0	0.011	0.137	0.784	0.063	0.005	0	0	0
6	0	0	0	0	0.011	0.114	0.795	0.080	0	0	0
7	0	0	0	0	0	0.003	0.188	0.760	0.035	0.014	0
8	0	0	0	0	0	0	0.020	0.149	0.741	0.090	0
9	0	0	0	0	0	0	0	0.013	0.161	0.719	0.098
10	0	0	0	0	0	0.007	0.010	0.013	0.066	0.185	0.629

A new exchange exercise has been carried out (ICES WGBIOP, report unpublished) from which an updated age error matrix is now available. Differences between both (Figure 16.23) appear to be small enough that the 2016 matrix is being kept in use in the model. No particular aging error was found with specimens from the 2018 year class in the most recent exchange.

16.7 Assessment

During the 2024 benchmark the assessment model changed from the statistical catch-at-age model *AAP* (Aarts & Poos 2009) to *Stock Synthesis 3* (SS3; Methot & Wetzel 2013). The SS3 reference model is age-based, single sex, and single area, with two fleets (landings and discards) and two abundance indices (*BTS* and *COAST*). Selectivity is time-varying (as a random walk) for landings and time-invariant for discards and both surveys, and all are set constant from age 5. Spawning-stock biomass (SSB) is calculated at the start of the year, and fecundity is assumed proportional to individual weight. Recruitment is assumed to take place at the beginning of the year, with 0 as recruitment age, and following a Beverton–Holt stock recruitment relationship. The steepness of the relationship has been fixed at a value of $h = 0.6646$ taken from *FishLife* correlations for *S. solea* (Thorson 2019). Ages 2–6 are treated as fully selected when calculating the mean fishing-mortality metric F' .

Sole in Subarea 4. Settings of the SS3 stock assessment model.

Setting	Description
Plus group	Age 10
First tuning year	1990
Selectivity for landings and discards constant for age	≥ 5
Selectivity for landings	Time-varying (random walk)
Selectivity for discards and survey	Time-invariant
Catchability surveys constant for ages	≥ 5

A summary of the assessment results (recruitment, F and SSB, including confidence bounds) is presented in Figure 16.24. Fishing pressure in 2024 is estimated to be the lowest in the series, while SSB is back to a level last seen in 1999. Recruitment fluctuates around its long-term mean but has been lower since 2019. The estimates of spawning stock biomass and corresponding recruitment at age 0 are shown in Figure 16.25. The proportion of spawning stock biomass estimated by age and year is presented in Figure 16.26. The strong 2018 year class has dominated the spawning stock since reaching maturity in 2021 (Figure 16.27), and still account for 32% for the spawning biomass in 2024. Catches of this cohort in 2024, as age 6 fish, accounted for 26% of the catch in weight (Figure 16.28).

Log-standardized residuals of the model fit to the four data sources employed (landings and discards numbers-at-age and the two indices of abundance) are presented in Figure 16.29. Some residual patterns are present in the model output, especially for the recent years in the BTS fleet, which could be linked to the estimates of fleet selectivity. Moreover, there is a cohort pattern in the residuals of the landings (Fleet) and age 4 is always overestimated, which has a significant contribution to the catches (Table 16.2).

One step ahead (OSA) residuals (Trijoulet, 2023) of the model fit to the four data sources employed (landings and discards numbers-at-age and the two indices of abundance) are presented in Figure 16.30. The one-step ahead residuals also show residual patterns for the BTS fleet, in most recent years and for ages 0–5. The runs test for the mean age in landings, discards and both indices (Carvalho *et al.*, 2021) are presented in Figure 16.31.

Fits to the BTS and COAST index are presented in Figures 16.32 and 16.33. The fit to the biomass trends in both indices of abundance is reasonable.

The selectivity of the fleet as estimated by the SS3 model is presented in Figure 16.34. The changes in pattern on landings selectivity in recent years appear to be linked to the progression of the strong 2018 year class. Stock numbers at age 4 are high in 2022 (Table 16.10) and have only been estimated as high 4 times over the entire time series. If a similar selectivity pattern (without drop at age 4) as estimated in the previous years would be applied, this would result in a much too high prediction of the numbers at age 4. Therefore, the model reduced the selectivity at age 4 (unexpected drop; ~40% reduction) to obtain a better fit. Despite this reduction, a high positive residual can still be observed for age 4 (Figure 16.29). Similarly, selectivity is estimated lower for age 5 (and older ages, as it is set constant; drop of ~35%) in 2024 compared to the previous years, where selectivity was always highest for these older ages. A similar pattern was repeated last year for the 2023 estimates. This requires some investigation on what factors are affecting the model estimate in the last available year.

The retrospective patterns for recruitment, spawning biomass and fishing mortality are summarized in Figure 16.35. The retrospective analysis for the model shows a robust configuration that does not suffer from a problematic Mohn’s rho value, being between 0.20 and –0.15 for both SSB and F ($\rho_{SSB} = 0.03$ and $\rho_F = -0.05$). Figures 16.36 and 16.37 present the results of an analysis of prediction skills by means of hindcasting cross-validation (carried out following Carvalho *et al.*, 2021). This diagnostic uses a series of five retrospective runs from which a one year forecast for the reported total catch is carried out. The predicted overall abundance index is then compared with the observed values. MASE (Mean Absolute Scaled Error) values reported in each figure indicate the model ability to predict those indices, better than a random walk when $MASE < 1$. The updated model run returns MASE values < 1 for the COAST index, but still larger than one for BTS.

Yearly estimates of abundances and fishing mortality-at-age obtained by the model run are presented in Tables 16.10 and 16.11, respectively.

16.8 Recruitment assumption

The short-term forecast for the stock requires an assumption about recruitment in the intermediate year, 2025. Following the benchmark logic, recruitment is set to the geometric mean of the 1957–2023 time-series of recruitment estimates, that is 1 425 229 thousand individuals. The most-recent recruit estimate (2024) was excluded because no recruitment deviance is estimated for the last year by the assessment model.

16.9 Short-term forecast and advice for 2026

Short-term forecasts were carried out with the following settings:

- Natural mortality and maturity for 2025-2026 set as the average of the last three years (2022-2024)
- Selectivity-at-age, and weights-at-age in landings, discards and stock for 2025-2026 set as the average of the last five years (2020-2024).
- Ratio of discards to landings at age as the average over the last three years (2022-2024).

Population numbers in the intermediate year for ages 1 and older are taken from the SS3 survivors estimates.

An overview of the interim year assumptions is given in the table below.

Sole in Subarea 4. Values in the forecast and for the interim year.

Variable	Value	Notes
F _{ages 2–6} (2025)	0.077	Based on catch constraint for 2025. Average exploitation pattern (2020–2024).
Spawning-stock biomass (SSB; 2025)	57 728	Short-term forecast (STF); tonnes
SSB (2026)	53 114	STF; tonnes
R _{age 0} (2025, 2026)	1 425 136	Geometric mean of recruitment (GM; 1957–2023); thousands
Total catch (2025)	5000	50% of the total allowable catch (TAC) for 2025; tonnes

Variable	Value	Notes
Projected landings (2025)	4486	Short-term forecast (STF), assuming average estimated landing ratio by age, 2022–2024; tonnes
Projected discards (2025)	514	STF, assuming average estimated discard ratio by age, 2022–2024; tonnes

Forecasts use the *FLR/FLasher* toolset (Kell *et al.* 2007; Scott & Mosqueira 2016). Source code is available in the corresponding TAF repository.

Intermediate-year catches are assumed to equal 50 % of the 2025 TAC. A large decreased in capacity in the main fleet targeting this stock (TBB 70-99) means total removals are likely to stay below the 10 000 t TAC. Fishing mortality in 2024 was exceptionally low due to the reduced bias-corrected advice and TAC so it does not provide a realistic assumption onm fishing pressure in 2025. Analysis of quarterly catches in 2023 (when a similar TAC of 9 152 t was in force) and the uptake reported over the first quarter of 2025, suggest a similar seasonal pattern in catch in those years. Both of these factors indicate that a 50% uptake is a more realistic assumption than either full TAC or status quo F.

The resulting catch advice for 2026 is presented in Table 16.12 and amounts to 12454 tonnes. The change in advice (+16.4%) is mainly due to increased weight-at-age observed in 2024.

16.10 Reference points

The precautionary and MSY reference points were updated at during WGNSSK in 2024, following the ICES WKMSYREF3 protocol. All values come from the SS3 reference run fitted with a segmented (hockey-stick) stock–recruit relationship. The reference points in use for the stock are as follows:

Sole in Subarea 4. Proposed set of reference points computed during the 2024 benchmark based on a segmented regression stock-recruitment relationship.

Framework	Reference point	Value	Technical basis	Source
Maximum sustainable yield (MSY) approach	MSY B _{trigger}	48 340	B _{PA} ; in tonnes	ICES (2024a)
	F _{MSY}	0.186	Stochastic simulations (EqSim) assuming a hockey stick stock–recruit relationship, based on the recruitment period 1958–2021. Capped at F _{PA} .	ICES (2024a)
Precautionary approach	B _{lim}	34 788	Lowest spawning-stock biomass (SSB) level that in the past has resulted in good recruitment; in tonnes	ICES (2024a)
	B _{PA}	48 340	B _{lim} × exp (1.645 × σ), σ = 0.20; in tonnes	ICES (2024a)
	F _{PA}	0.186	The F that provides a 95% probability for SSB to be above B _{lim} (F _{P05} with advice rule)	ICES (2024a)
EU management plan (MAP)*	MAP MSY B _{trigger}	48 340	MSY B _{trigger} ; in tonnes	ICES (2024a)

Framework	Reference point	Value	Technical basis	Source
	MAP B_{lim}	34 788	B_{lim} ; in tonnes	ICES (2024a)
	MAP F_{MSY}	0.186	F_{MSY}	ICES (2024a)
	MAP range F_{lower}	0.168–0.186	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY	ICES (2024a)
	MAP range F_{upper}	0.186–0.186	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY	ICES (2024a)

*EU multiannual plan (MAP) for the North Sea (EU, 2018).

16.11 Quality of the assessment

The 2024 benchmark introduced SS3, revised survey indices, updated catch statistics, and a size-based natural-mortality assumption. Fit to the survey biomass indices has improved for the 2025 update, yet changes patterns in abundance in 2019 and 2020 are not well explained. Potential causes, such as changes in spatial availability to the different surveys, are not fully understood, but work is ongoing to explain them.

16.12 Status of the stock

The current assessment estimates a substantial increase in SSB in 2024 following the reduction in catches but also largely driven by the increase in the computed mean weights-at-age in stock (Figure 16.9). This increase is largely consistent across ages, so could result from favourable conditions for the stock, although observation error could also have influenced them. Recruitment over the last three years is estimated to have oscillated from above the long term average in 2022 to half of that in 2023. The current assessment estimates spawning stock biomass to be above the B_{lim} and $MSYB_{trigger}$ values, and fishing pressure is below F_{MSY} .

16.13 Management considerations

Sole is mostly taken by beam trawlers in a mixed fishery with plaice in the southern and part of the North Sea. The fishery has changed notably over the last decade with the adoption and subsequent ban of pulse fishing, which was especially successful catching sole.

Discards of sole are allowed for multiple gears under the survivability and *de minimis* exemptions. These include the beam trawlers (TBB-DEF-70-99) targeting the stock, and responsible for most of the catch.

16.14 Issues for future benchmark

- With the move to Stock Synthesis in its current formulation, with a separate discards fleet to allow different mean weights-at-age, estimates of past discards at age are not available. This could have implications on stock productivity estimates, although previous values like those returned by AAP were not higher than current. Modelling of discards prior to

2002 could be attempted based on available discard ratios and their relationship with total catch, cohort strength and even economic factors.

- The observed short-term variability in mean weights-at-age in the catch, also employed as values for those in the stock is probably in a large proportion the result of observation error, especially for older ages for which relatively few individuals are sampled. The effect of these changes in the estimated SSB for 2024 has been especially large. Although this recent increase in weights is observed across a range of ages, which makes it more likely to reflect real changes in the stock, other values in recent years need further investigation. Modelling growth is an option that should be explored inter-sessionally.
- The ability of both indices of abundance to monitor the development of the 2018 year class is still being questioned. Changes in the spatial distribution of the younger ages, as a result of either or both density dependence and climate change, appear like a possible explanation for some of the patterns observed. Large differences have also been observed in the trends for certain cohorts between the standardized BTS and the mean density calculation. This discrepancy could be caused by some of the settings in the GAM survey model. Work is ongoing at WMR to clarify this point.
- The maturity assumption was not updated during the last benchmark, and remains as a knife-edge ogive. The available data on maturity is limited and comes in a great part from surveys carried out in quarter 3, too late in the year. A recommendation should be made for the collection of maturity data on sole during quarters 1 and 2.
- Finally, some of the issue above might be driven by the spatial distribution of the stock not being as homogenous as assumed on a single-stock assessment model. Patterns in abundance appear to differ between the Western and Eastern sides of the area of distribution of the stock, as suggested by both surveys and fishery data.

16.15 References

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16.16 Figures and tables

Table 16.1. Sole in Subarea 4. Time-series of the official landings by country and overall total, the official BMS landings, the landings and discards reported to ICES and the total TAC (figures rounded to the nearest tonne).

Year	Belgium	Denmark	France	Germany	Netherlands	UK	Other	Official landings	Official BMS landings	ICES landings	ICES discards	TAC
1982	1900	524	686	266	17686	403	2	21467		21578		21000
1983	1740	730	332	619	16101	435	0	19957		24927		20000
1984	1771	818	400	1034	14330	586	1	18940		26839		20000
1985	2390	692	875	303	14897	774	3	19934		24248		22000
1986	1833	443	296	155	9558	647	2	12934		18201		20000
1987	1644	342	318	210	10635	676	4	13829		17368		14000
1988	1199	616	487	452	9841	740	28	13363		21590		14000
1989	1596	1020	312	864	9620	1033	50	14495		21804		14000
1990	2389	1427	352	2296	18202	1614	263	26543		35121		25000
1991	2977	1307	465	2107	18758	1723	271	27608		33514		27000
1992	2058	1359	548	1880	18601	1281	277	26004		29341		25000
1993	2783	1661	490	1379	22015	1149	298	29775		31491		32000
1994	2935	1804	499	1744	22874	1137	298	31291		33002		32000
1995	2624	1673	640	1564	20927	1040	312	28780		30468		28000
1996	2555	1018	535	670	15344	848	229	21199		22650		23000
1997	1519	689	99	510	10241	479	204	13741		14902		18000
1998	1844	520	510	782	15198	549	339	19742		20867		19100
1999	1919	828	n/a	1458	16283	645	501	21634*		23475		22000
2000	1806	1069	362	1280	15273	600	539	20929		22641		22000
2001	1874	772	411	958	13345	597	394	18351		19944		19000
2002	1437	644	266	759	12120	451	292	15969		15944	3457	16000
2003	1605	703	728	749	12469	521	363	17138		15654	1902	15900
2004	1477	808	655	949	12860	535	544	17828		17932	3281	17000
2005	1374	831	676	756	10917	667	357	15579		15777	1885	18600
2006	986	582	713	475	8299	912	36	12003		12216	1986	17700

Year	Belgium	Denmark	France	Germany	Netherlands	UK	Other	Official landings	Official BMS landings	ICES landings	ICES dis-cards	TAC
2007	972	413	591	458	10364	1207	5	14010		14439	1406	15000
2008	1379	501	574	513	9456	851	16	13289		14206	897	12800
2009	1367	476	909	554	9606	951	1	13864		14282	1210	14000
2010	1267	405	630	537	8770	942	2	12552		12793	2896	14100
2011	864	346	566	327	8137	820	2	11061		11497	1863	14100
2012	607	418	633	440	9085	609	3	11794		12210	2854	16200
2013	706	497	687	560	9967	870	1	13287		13145	2336	14000
2014	966	313	674	642	9018	842	<0.5	12455		13091	1949	11900
2015	934	270	541	764	9273	812	<0.5	12594		12920	1755	11900
2016	767	354	362	861	9600	705	<0.5	12649		14133	1237	13262
2017	556	433	392	760	9482	513	<0.5	12136	30	12373	1325	16123
2018	403	368	431	721	8580	431	3	10936	57	11193	1184	15694
2019	253	109	109	620	6997	333	1	8422	47	8647	1963	12555
2020	240	123	36	919	6759	547	<0.5	8624	43	8864	1472	17545
2021	270	172	166	648	6231	476	<0.5	7963	36	8238	955	21361
2022	165	77	25	144	4316	348	1	5076	22	5291	606	15330
2023	158	42	4	190	3196	282	<0.5	3873	21	4029	386	9152
2024**	239	25	3	261	2924	323	<0.5	3775	17	3960	395	3675

* Total does not include official landings of all countries.

** Preliminary.

n/a = not available.

Table 16.2. Sole in Subarea 4. Time-series of landings at age (in thousands) which serves as input to SS3.

Year	1	2	3	4	5	6	7	8	9	10+
1957	0	1472	10556	13150	3913	3041	6780	1803	529	6541
1958	0	1863	8482	14240	9547	3501	3023	4461	2264	6590
1959	0	3694	12139	10499	9060	5823	1217	2044	2598	5668
1960	0	11965	14043	16691	9248	8313	4815	1583	1049	7851
1961	0	972	50470	19403	12574	4760	3998	4338	847	7355

Year	1	2	3	4	5	6	7	8	9	10+
1962	0	1584	6173	58836	15254	10478	4797	4087	2074	7450
1963	0	670	8271	8485	45823	8420	6603	2403	3365	8316
1964	53	150	2041	5518	3680	16749	3020	1749	790	2913
1965	0	45180	1045	1534	4798	2381	11990	1494	1463	3077
1966	0	12145	132170	979	1168	3649	736	6255	694	2424
1967	0	3769	26260	87039	1998	548	1962	777	5160	2978
1968	1034	17093	13852	24894	48417	461	244	1639	323	6502
1969	404	24404	21884	5433	12638	25646	338	249	1214	5379
1970	1299	6141	25996	8236	1784	3231	11961	246	140	5234
1971	425	33765	14596	12909	4538	1459	2355	7300	194	4649
1972	354	7511	36356	6997	4911	1548	517	1218	4654	2772
1973	716	12459	13025	16493	4101	2368	1013	779	1241	5899
1974	100	15171	21248	5412	6965	1896	1563	649	396	4750
1975	267	23193	28833	11839	2110	3870	798	916	513	3481
1976	1064	3619	28571	14316	4923	987	1950	562	434	2721
1977	1780	22747	12299	15593	7580	1812	325	1133	261	2155
1978	27	24921	29163	6102	6610	4231	1730	608	643	1595
1979	9	8280	41681	16259	3033	3262	1769	826	244	1546
1980	650	1233	12762	18138	7444	1479	2241	1437	374	1227
1981	434	29983	3344	7046	8439	3757	973	909	786	932
1982	2697	26799	46375	1868	3584	4855	1701	623	613	1295
1983	391	34545	41551	21273	626	1383	1958	982	388	1181
1984	192	30839	44081	22631	8821	744	857	1047	526	897
1985	163	16449	42773	20079	9307	3520	207	375	631	965
1986	372	9304	18381	17591	7698	5480	2256	109	281	1671
1987	93	28896	21927	8851	6477	3102	1559	898	81	690
1988	10	13206	47135	15217	4377	3878	1549	890	523	317
1989	115	45652	17973	22295	4551	1627	1414	637	451	459
1990	854	11816	103380	9667	9099	3315	1032	1186	548	837

Year	1	2	3	4	5	6	7	8	9	10+
1991	118	12938	24985	76580	6609	3612	1706	707	718	1072
1992	965	6730	43713	15961	37745	2440	2995	730	393	1163
1993	53	49870	16575	31047	13709	23758	1472	1170	456	833
1994	709	7710	86349	13387	18513	5642	11174	458	905	897
1995	4766	12674	16700	68073	6262	7254	1981	5971	293	665
1996	170	18609	16005	16770	26946	3814	4725	932	3267	976
1997	1574	5987	23418	7253	5058	12667	1189	2303	330	1672
1998	242	56162	15011	14806	3466	1924	4727	787	1022	838
1999	284	15601	71730	8103	6049	1200	657	1964	328	804
2000	2329	14929	32425	42394	3257	2453	796	431	922	708
2001	857	25045	20925	19260	16211	1383	808	266	163	701
2002	1098	10963	31237	10090	7017	6064	599	470	86	291
2003	849	29423	13902	13746	4654	3335	2442	322	90	450
2004	255	14911	45860	8245	7007	2100	872	860	339	538
2005	1182	7014	22271	28499	3831	3781	1325	566	683	336
2006	4675	9741	11480	10343	11944	1429	1592	773	380	496
2007	298	39680	9713	6275	5578	4819	715	645	432	446
2008	2344	6827	37763	6076	2889	2262	2328	343	363	410
2009	1655	11588	11403	25748	3249	1695	1539	1475	256	832
2010	424	11809	13979	8158	13588	1884	1092	803	1140	993
2011	6	12121	19959	8758	5028	6364	1020	459	317	893
2012	0	6665	27450	10878	4501	1930	3956	589	261	1067
2013	0	2480	25770	19556	5896	2316	1630	1748	335	1114
2014	452	8025	6713	21576	11343	3118	1830	806	880	990
2015	212	10703	15618	4789	14299	6187	1629	652	513	904
2016	120	6256	24006	11386	4704	8894	4523	904	322	1143
2017	273	4917	16263	15816	5827	2044	3812	1949	488	565
2018	375	11836	9255	11696	9935	3751	991	1835	773	386
2019	457	6366	15708	6220	5030	4659	1544	695	1630	423

Year	1	2	3	4	5	6	7	8	9	10+
2020	131	17752	9891	10600	2684	2231	1442	596	158	1301
2021	340	2293	24628	7587	4415	1104	687	704	225	127
2022	291	3043	2568	13061	2563	1860	629	337	250	85
2023	1341	3975	3486	1455	4910	1062	696	163	123	147
2024	242	5870	4869	2366	688	2413	626	291	77	96

Table 16.3. Sole in Subarea 4. Time-series of discards at age (in thousands) which serves as input to SS3.

Year	1	2	3	4	5	6	7	8	9	10+
2002	10963	21132	8807	1750	463	0	0	0	0	0
2003	535	15951	2063	418	47	39	9	0	0	0
2004	4314	19030	11369	2522	758	17	0	0	0	0
2005	3584	8486	6950	1796	540	77	27	0	0	0
2006	5091	10907	5018	1192	362	44	30	12	0	0
2007	4774	8366	4810	998	341	80	6	1	1	0
2008	1939	4516	2703	576	170	15	10	0	0	0
2009	3238	5606	861	1499	289	48	21	0	0	0
2010	5631	11875	8945	1954	2442	111	1623	21	76	0
2011	3581	13750	3658	1550	105	96	2	0	0	0
2012	1198	22305	10458	915	239	49	15	0	8	0
2013	4587	6387	14095	3975	326	64	14	1	0	0
2014	20638	6959	2746	4023	1556	256	229	7	0	0
2015	5013	8777	3890	1976	1545	372	15	49	8	1
2016	4309	5327	5286	1466	297	567	248	3	0	0
2017	5734	4462	3968	2949	494	88	368	143	0	0
2018	2850	8791	1858	1588	1267	116	16	141	39	0
2019	12178	8530	5641	1469	819	820	249	20	84	1
2020	2531	13753	2270	1251	337	74	66	13	6	0
2021	2375	1528	5807	924	395	39	9	20	6	0
2022	1401	2601	711	2219	296	215	26	31	25	0
2023	1315	1719	632	175	598	89	4	0	8	0

Year	1	2	3	4	5	6	7	8	9	10+
2024	1188	2155	720	95	23	104	16	1	1	0

Table 16.4. Sole in Subarea 4. Time-series of the mean weights-at-age in the landings of sole in Subarea 27.4. Age 0 is 0.01 constant for all years. Input to SS3.

Year	1	2	3	4	5	6	7	8	9	10+
1957	0.155	0.154	0.177	0.204	0.248	0.279	0.290	0.335	0.436	0.408
1958	0.155	0.145	0.178	0.220	0.254	0.273	0.314	0.323	0.388	0.413
1959	0.155	0.162	0.188	0.228	0.261	0.301	0.328	0.321	0.373	0.426
1960	0.155	0.153	0.185	0.235	0.254	0.277	0.301	0.309	0.381	0.418
1961	0.155	0.146	0.174	0.211	0.255	0.288	0.319	0.304	0.346	0.419
1962	0.155	0.155	0.165	0.208	0.241	0.295	0.320	0.321	0.334	0.412
1963	0.155	0.163	0.171	0.219	0.258	0.309	0.323	0.387	0.376	0.485
1964	0.153	0.175	0.213	0.252	0.274	0.309	0.327	0.346	0.388	0.481
1965	0.155	0.169	0.209	0.246	0.286	0.282	0.345	0.378	0.404	0.480
1966	0.155	0.177	0.190	0.180	0.301	0.332	0.429	0.399	0.449	0.501
1967	0.155	0.192	0.201	0.252	0.277	0.389	0.419	0.339	0.424	0.491
1968	0.157	0.189	0.207	0.267	0.327	0.342	0.354	0.455	0.465	0.508
1969	0.152	0.191	0.196	0.255	0.311	0.373	0.553	0.398	0.468	0.523
1970	0.154	0.212	0.218	0.285	0.350	0.404	0.441	0.463	0.443	0.533
1971	0.145	0.193	0.237	0.322	0.358	0.425	0.420	0.490	0.534	0.547
1972	0.169	0.204	0.252	0.334	0.434	0.425	0.532	0.485	0.558	0.629
1973	0.146	0.208	0.238	0.346	0.404	0.448	0.552	0.567	0.509	0.586
1974	0.164	0.192	0.233	0.338	0.418	0.448	0.520	0.559	0.609	0.653
1975	0.129	0.182	0.225	0.320	0.406	0.456	0.529	0.595	0.629	0.669
1976	0.143	0.190	0.222	0.306	0.389	0.441	0.512	0.562	0.667	0.665
1977	0.147	0.188	0.236	0.307	0.369	0.424	0.430	0.520	0.562	0.619
1978	0.152	0.196	0.231	0.314	0.370	0.426	0.466	0.417	0.572	0.666
1979	0.137	0.208	0.246	0.323	0.391	0.448	0.534	0.544	0.609	0.763
1980	0.141	0.199	0.244	0.331	0.371	0.418	0.499	0.550	0.598	0.684
1981	0.143	0.187	0.226	0.324	0.378	0.424	0.442	0.516	0.542	0.630
1982	0.141	0.188	0.216	0.307	0.371	0.409	0.437	0.491	0.580	0.656

Year	1	2	3	4	5	6	7	8	9	10+
1983	0.134	0.182	0.217	0.301	0.389	0.416	0.467	0.489	0.505	0.642
1984	0.153	0.171	0.221	0.286	0.361	0.386	0.465	0.555	0.575	0.634
1985	0.122	0.187	0.216	0.288	0.357	0.427	0.447	0.544	0.612	0.645
1986	0.135	0.179	0.213	0.299	0.357	0.407	0.485	0.543	0.568	0.610
1987	0.139	0.185	0.205	0.277	0.356	0.378	0.428	0.481	0.393	0.657
1988	0.127	0.175	0.217	0.270	0.354	0.428	0.484	0.521	0.559	0.712
1989	0.118	0.173	0.216	0.288	0.336	0.375	0.456	0.492	0.470	0.611
1990	0.124	0.183	0.227	0.292	0.371	0.413	0.415	0.514	0.476	0.620
1991	0.127	0.186	0.210	0.263	0.315	0.436	0.443	0.467	0.507	0.558
1992	0.146	0.178	0.213	0.258	0.298	0.380	0.409	0.460	0.487	0.556
1993	0.097	0.167	0.196	0.239	0.264	0.300	0.338	0.441	0.496	0.603
1994	0.143	0.180	0.202	0.228	0.257	0.300	0.317	0.432	0.409	0.510
1995	0.151	0.186	0.196	0.247	0.265	0.319	0.344	0.356	0.444	0.592
1996	0.163	0.177	0.202	0.234	0.274	0.285	0.318	0.370	0.390	0.594
1997	0.151	0.180	0.206	0.236	0.267	0.296	0.323	0.306	0.384	0.440
1998	0.128	0.182	0.189	0.252	0.262	0.289	0.336	0.292	0.335	0.504
1999	0.163	0.179	0.212	0.229	0.287	0.324	0.354	0.372	0.372	0.453
2000	0.145	0.170	0.200	0.248	0.290	0.299	0.323	0.368	0.402	0.428
2001	0.143	0.185	0.202	0.270	0.275	0.333	0.391	0.414	0.433	0.493
2002	0.142	0.182	0.218	0.254	0.278	0.309	0.356	0.332	0.583	0.604
2003	0.140	0.188	0.220	0.258	0.258	0.305	0.346	0.330	0.555	0.398
2004	0.138	0.183	0.209	0.252	0.269	0.285	0.388	0.376	0.332	0.462
2005	0.168	0.186	0.204	0.240	0.254	0.272	0.270	0.331	0.320	0.373
2006	0.152	0.177	0.210	0.248	0.277	0.313	0.281	0.340	0.390	0.385
2007	0.152	0.184	0.210	0.241	0.260	0.287	0.321	0.304	0.299	0.343
2008	0.151	0.181	0.223	0.243	0.278	0.333	0.320	0.333	0.352	0.427
2009	0.152	0.190	0.211	0.258	0.275	0.279	0.329	0.336	0.331	0.393
2010	0.160	0.179	0.216	0.237	0.272	0.302	0.294	0.316	0.379	0.390
2011	0.144	0.160	0.191	0.228	0.239	0.271	0.266	0.288	0.326	0.353

Year	1	2	3	4	5	6	7	8	9	10+
2012	0.144	0.169	0.189	0.234	0.255	0.252	0.278	0.295	0.340	0.332
2013	0.145	0.169	0.185	0.222	0.248	0.263	0.294	0.316	0.319	0.365
2014	0.138	0.190	0.213	0.230	0.261	0.268	0.257	0.274	0.321	0.345
2015	0.146	0.170	0.209	0.243	0.264	0.273	0.299	0.298	0.328	0.363
2016	0.150	0.173	0.199	0.234	0.256	0.272	0.271	0.288	0.341	0.325
2017	0.152	0.177	0.205	0.240	0.282	0.275	0.312	0.310	0.279	0.347
2018	0.123	0.166	0.200	0.235	0.258	0.265	0.241	0.260	0.268	0.265
2019	0.156	0.163	0.186	0.224	0.234	0.228	0.243	0.219	0.197	0.319
2020	0.147	0.158	0.184	0.214	0.236	0.236	0.214	0.233	0.210	0.203
2021	0.133	0.166	0.181	0.206	0.236	0.257	0.271	0.236	0.210	0.381
2022	0.127	0.159	0.186	0.223	0.238	0.239	0.245	0.264	0.199	0.383
2023	0.160	0.190	0.217	0.233	0.276	0.265	0.303	0.271	0.292	0.245
2024	0.140	0.186	0.219	0.249	0.263	0.289	0.267	0.267	0.319	0.306

Table 16.5. Sole in Subarea 4. Time-series of the mean weights-at-age in the discards of sole in Subarea 27.4. Input to SS3.

Year	0	1	2	3	4	5	6	7	8	9	10+
1957	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1958	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1959	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1960	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1961	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1962	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1963	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1964	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1965	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1966	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1967	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1968	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1969	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1970	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137

Year	0	1	2	3	4	5	6	7	8	9	10+
1971	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1972	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1973	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1974	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1975	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1976	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1977	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1978	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1979	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1980	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1981	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1982	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1983	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1984	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1985	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1986	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1987	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1988	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1989	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1990	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1991	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1992	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1993	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1994	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1995	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1996	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1997	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1998	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
1999	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137

Year	0	1	2	3	4	5	6	7	8	9	10+
2000	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
2001	0.010	0.059	0.082	0.096	0.104	0.112	0.106	0.120	0.131	0.137	0.137
2002	0.010	0.070	0.080	0.089	0.096	0.101	0.086	0.085	0.088	0.137	0.137
2003	0.010	0.036	0.099	0.116	0.115	0.157	0.161	0.186	0.166	0.325	0.321
2004	0.010	0.067	0.080	0.097	0.108	0.118	0.096	0.101	0.098	0.101	0.321
2005	0.010	0.051	0.086	0.098	0.107	0.115	0.103	0.107	0.098	0.101	0.321
2006	0.010	0.074	0.086	0.098	0.105	0.109	0.115	0.113	0.121	0.113	0.117
2007	0.010	0.026	0.081	0.091	0.098	0.105	0.104	0.116	0.102	0.102	0.111
2008	0.010	0.074	0.085	0.098	0.111	0.132	0.121	0.110	0.091	0.100	0.090
2009	0.010	0.068	0.101	0.106	0.107	0.109	0.145	0.127	0.092	0.100	0.086
2010	0.010	0.052	0.088	0.095	0.115	0.117	0.138	0.097	0.190	0.142	0.118
2011	0.010	0.051	0.083	0.094	0.090	0.134	0.128	0.177	0.153	0.199	0.182
2012	0.010	0.062	0.075	0.090	0.101	0.111	0.113	0.123	0.095	0.121	0.095
2013	0.010	0.043	0.075	0.086	0.100	0.121	0.110	0.108	0.136	0.140	0.100
2014	0.010	0.026	0.079	0.089	0.097	0.106	0.101	0.117	0.099	0.155	0.106
2015	0.010	0.057	0.077	0.095	0.087	0.105	0.124	0.131	0.124	0.159	0.200
2016	0.010	0.029	0.073	0.086	0.094	0.113	0.105	0.119	0.203	0.207	0.180
2017	0.010	0.051	0.073	0.086	0.085	0.093	0.109	0.106	0.101	0.123	0.161
2018	0.010	0.041	0.066	0.086	0.091	0.097	0.102	0.100	0.105	0.126	0.087
2019	0.010	0.045	0.071	0.133	0.153	0.191	0.168	0.177	0.224	0.194	0.248
2020	0.010	0.053	0.071	0.144	0.174	0.218	0.214	0.197	0.249	0.173	0.244
2021	0.010	0.070	0.080	0.089	0.101	0.111	0.117	0.120	0.099	0.138	0.138
2022	0.010	0.066	0.068	0.085	0.096	0.103	0.113	0.110	0.089	0.112	0.138
2023	0.010	0.072	0.085	0.087	0.106	0.098	0.122	0.114	0.169	0.169	0.169
2024	0.010	0.064	0.088	0.099	0.107	0.122	0.114	0.118	0.117	0.149	0.149

Table 16.6. Sole in Subarea 4. Time-series of stock weights-at-age in the middle of the year. Input to SS3.

Year	0	1	2	3	4	5	6	7	8	9	10+
1957	0.010	0.025	0.070	0.147	0.187	0.208	0.253	0.262	0.355	0.390	0.365
1958	0.010	0.025	0.070	0.164	0.205	0.226	0.228	0.297	0.318	0.393	0.422

Year	0	1	2	3	4	5	6	7	8	9	10+
1959	0.010	0.025	0.070	0.159	0.198	0.239	0.271	0.292	0.276	0.303	0.426
1960	0.010	0.025	0.070	0.163	0.207	0.234	0.240	0.268	0.242	0.360	0.431
1961	0.010	0.025	0.070	0.148	0.206	0.235	0.232	0.259	0.274	0.281	0.396
1962	0.010	0.025	0.070	0.148	0.192	0.240	0.301	0.293	0.282	0.273	0.441
1963	0.010	0.025	0.070	0.148	0.193	0.243	0.275	0.311	0.363	0.329	0.465
1964	0.010	0.025	0.070	0.159	0.214	0.240	0.291	0.305	0.306	0.365	0.474
1965	0.010	0.025	0.140	0.198	0.223	0.251	0.297	0.337	0.358	0.526	0.460
1966	0.010	0.025	0.070	0.160	0.149	0.389	0.310	0.406	0.377	0.385	0.505
1967	0.010	0.025	0.177	0.164	0.235	0.242	0.399	0.362	0.283	0.381	0.459
1968	0.010	0.025	0.122	0.171	0.248	0.312	0.280	0.629	0.416	0.410	0.486
1969	0.010	0.025	0.137	0.174	0.252	0.324	0.364	0.579	0.415	0.469	0.521
1970	0.010	0.025	0.137	0.201	0.275	0.341	0.367	0.423	0.458	0.390	0.554
1971	0.010	0.034	0.148	0.213	0.313	0.361	0.410	0.432	0.474	0.483	0.533
1972	0.010	0.038	0.155	0.218	0.313	0.419	0.443	0.443	0.443	0.508	0.602
1973	0.010	0.039	0.149	0.226	0.322	0.371	0.433	0.452	0.472	0.446	0.536
1974	0.010	0.035	0.146	0.218	0.329	0.408	0.429	0.499	0.565	0.542	0.618
1975	0.010	0.035	0.148	0.206	0.311	0.403	0.446	0.508	0.582	0.580	0.650
1976	0.010	0.035	0.142	0.201	0.301	0.379	0.458	0.508	0.517	0.644	0.665
1977	0.010	0.035	0.147	0.202	0.291	0.365	0.409	0.478	0.487	0.531	0.644
1978	0.010	0.035	0.139	0.211	0.290	0.365	0.429	0.427	0.385	0.542	0.644
1979	0.010	0.045	0.148	0.211	0.300	0.352	0.429	0.521	0.562	0.567	0.743
1980	0.010	0.039	0.157	0.200	0.304	0.345	0.394	0.489	0.537	0.579	0.645
1981	0.010	0.050	0.137	0.200	0.305	0.364	0.402	0.454	0.522	0.561	0.622
1982	0.010	0.050	0.130	0.193	0.270	0.359	0.411	0.429	0.476	0.583	0.642
1983	0.010	0.050	0.140	0.200	0.285	0.329	0.435	0.464	0.483	0.510	0.636
1984	0.010	0.050	0.133	0.203	0.268	0.348	0.386	0.488	0.591	0.567	0.663
1985	0.010	0.050	0.127	0.185	0.267	0.324	0.381	0.380	0.626	0.554	0.642
1986	0.010	0.050	0.133	0.191	0.278	0.345	0.423	0.495	0.487	0.587	0.686
1987	0.010	0.050	0.154	0.191	0.262	0.357	0.381	0.406	0.454	0.332	0.620

Year	0	1	2	3	4	5	6	7	8	9	10+
1988	0.010	0.050	0.133	0.193	0.260	0.335	0.409	0.417	0.474	0.486	0.654
1989	0.010	0.050	0.133	0.195	0.290	0.350	0.340	0.411	0.475	0.419	0.594
1990	0.010	0.050	0.148	0.203	0.294	0.357	0.447	0.399	0.494	0.481	0.653
1991	0.010	0.050	0.139	0.184	0.254	0.301	0.413	0.447	0.522	0.548	0.573
1992	0.010	0.050	0.156	0.194	0.257	0.307	0.398	0.406	0.472	0.500	0.540
1993	0.010	0.050	0.128	0.184	0.229	0.265	0.293	0.344	0.482	0.437	0.583
1994	0.010	0.050	0.143	0.174	0.209	0.257	0.326	0.349	0.402	0.494	0.459
1995	0.010	0.050	0.151	0.179	0.240	0.253	0.321	0.365	0.357	0.545	0.545
1996	0.010	0.050	0.147	0.178	0.208	0.274	0.268	0.321	0.375	0.402	0.546
1997	0.010	0.050	0.150	0.190	0.225	0.252	0.303	0.319	0.325	0.360	0.424
1998	0.010	0.050	0.140	0.173	0.234	0.267	0.281	0.328	0.273	0.336	0.455
1999	0.010	0.050	0.131	0.187	0.216	0.259	0.296	0.340	0.322	0.369	0.464
2000	0.010	0.050	0.139	0.185	0.226	0.264	0.275	0.287	0.337	0.391	0.376
2001	0.010	0.050	0.144	0.185	0.223	0.263	0.319	0.327	0.421	0.410	0.530
2002	0.010	0.032	0.108	0.192	0.236	0.260	0.265	0.282	0.305	0.463	0.496
2003	0.010	0.028	0.110	0.191	0.226	0.234	0.273	0.304	0.325	0.533	0.413
2004	0.010	0.042	0.093	0.170	0.190	0.224	0.273	0.314	0.388	0.335	0.457
2005	0.010	0.046	0.086	0.159	0.221	0.214	0.251	0.231	0.447	0.360	0.367
2006	0.010	0.029	0.096	0.166	0.238	0.270	0.338	0.304	0.390	0.422	0.495
2007	0.010	0.024	0.120	0.137	0.203	0.248	0.250	0.318	0.257	0.306	0.402
2008	0.010	0.075	0.108	0.192	0.221	0.266	0.324	0.295	0.289	0.372	0.396
2009	0.010	0.075	0.122	0.203	0.230	0.247	0.290	0.286	0.323	0.311	0.368
2010	0.010	0.070	0.114	0.163	0.173	0.235	0.280	0.308	0.336	0.345	0.385
2011	0.010	0.017	0.087	0.166	0.209	0.260	0.286	0.346	0.362	0.510	0.411
2012	0.010	0.015	0.059	0.142	0.207	0.227	0.260	0.261	0.288	0.481	0.371
2013	0.010	0.032	0.068	0.117	0.182	0.247	0.251	0.300	0.289	0.314	0.388
2014	0.010	0.010	0.081	0.136	0.188	0.210	0.229	0.232	0.297	0.345	0.394
2015	0.010	0.049	0.074	0.142	0.148	0.226	0.242	0.254	0.297	0.370	0.365
2016	0.010	0.011	0.068	0.149	0.182	0.230	0.245	0.236	0.261	0.245	0.307

Year	0	1	2	3	4	5	6	7	8	9	10+
2017	0.010	0.035	0.078	0.148	0.189	0.235	0.241	0.246	0.217	0.235	0.300
2018	0.010	0.009	0.083	0.145	0.177	0.203	0.235	0.225	0.217	0.256	0.310
2019	0.010	0.029	0.073	0.133	0.157	0.192	0.173	0.176	0.222	0.191	0.246
2020	0.010	0.038	0.082	0.143	0.175	0.218	0.213	0.209	0.246	0.168	0.238
2021	0.010	0.060	0.099	0.135	0.183	0.196	0.228	0.282	0.173	0.194	0.412
2022	0.010	0.037	0.065	0.137	0.173	0.193	0.204	0.245	0.195	0.209	0.232
2023	0.010	0.048	0.112	0.163	0.201	0.221	0.240	0.244	0.266	0.226	0.210
2024	0.010	0.053	0.135	0.198	0.229	0.256	0.272	0.247	0.280	0.231	0.326

Table 16.7. Sole in Subarea 4. Maturity and natural mortality-at-age in the stock.

Age	Maturity	M
0	0	1.302
1	0.0	0.754
2	0.0	0.465
3	1.0	0.368
4	1.0	0.321
5	1.0	0.295
6	1.0	0.278
7	1.0	0.267
8	1.0	0.260
9	1.0	0.254
10+	1.0	0.248

Table 16.8a. Sole in Subarea 4. Index of abundance at age, based on the BTS Q3 survey samples from The Netherlands, Germany, Belgium and the United Kingdom, used in the assessment of sole in Subarea 27.4.

Year	0	1	2	3	4	5	6	7	8	9	10+
1990	378.8	5457.0	4070.0	5093.0	447.7	207.4	80.8	12.3	8.9	11.0	8.6
1991	1209.0	3846.0	4862.0	1169.0	2183.0	123.1	67.4	22.9	13.6	19.7	26.9
1992	310.8	22470.0	1834.0	2629.0	925.1	1858.0	41.9	65.5	7.3	5.5	9.2
1993	220.2	5301.0	9480.0	571.0	777.8	358.9	576.8	15.2	29.1	14.0	34.5
1994	599.9	6009.0	1460.0	3640.0	87.2	335.2	74.4	271.8	10.2	5.6	12.7

Year	0	1	2	3	4	5	6	7	8	9	10+
1995	537.4	6043.0	2304.0	1320.0	1885.0	320.8	287.8	69.3	186.3	16.9	18.3
1996	82.2	2147.0	3151.0	850.8	809.9	1017.0	472.1	378.1	74.1	170.8	83.6
1997	346.7	13850.0	1962.0	866.6	306.1	233.0	224.0	111.5	43.4	32.7	28.2
1998	158.2	3155.0	5363.0	538.7	379.5	135.6	104.9	167.0	72.7	13.4	50.0
1999	503.6	5109.0	2331.0	2476.0	404.6	360.2	72.1	29.3	166.5	72.2	122.4
2000	161.2	3951.0	1919.0	808.2	708.3	169.3	50.6	17.1	11.1	69.7	38.4
2001	273.4	2267.0	3116.0	1108.0	484.4	416.0	85.6	45.6	15.3	5.2	55.2
2002	319.9	4126.0	1163.0	872.6	559.3	140.4	221.4	51.0	13.8	14.7	81.4
2003	237.2	3044.0	2920.0	466.5	433.7	202.8	48.2	130.5	17.2	23.9	20.2
2004	167.6	1663.0	1604.0	1131.0	204.3	206.6	31.5	37.7	5.8	8.9	16.5
2005	471.1	1888.0	1319.0	576.9	728.8	119.1	171.9	85.7	12.1	17.9	58.9
2006	156.8	5159.0	1436.0	359.3	396.7	351.1	64.4	83.9	47.5	6.4	46.5
2007	309.5	1959.0	3737.0	548.7	184.2	216.3	254.4	44.9	49.9	26.6	25.3
2008	319.2	2964.0	1746.0	1759.0	378.8	88.9	96.6	163.0	21.3	34.8	30.2
2009	261.3	3573.0	1805.0	882.2	1024.0	193.5	52.5	111.8	93.0	46.9	74.2
2010	291.5	3949.0	1627.0	589.5	383.7	316.4	122.6	33.6	33.7	27.6	53.4
2011	326.1	4221.0	3597.0	824.2	306.1	264.7	224.3	44.3	21.9	26.6	49.4
2012	128.1	1799.0	4819.0	1908.0	466.1	202.4	141.5	92.1	41.0	21.2	53.3
2013	101.0	2006.0	1331.0	2678.0	923.7	237.4	58.4	85.3	82.5	13.8	47.5
2014	1256.0	4759.0	3027.0	692.5	1220.0	506.6	114.4	45.1	46.5	27.3	16.3
2015	155.7	4505.0	2993.0	1706.0	553.0	984.7	333.9	147.8	29.0	49.5	53.4
2016	394.1	1911.0	2971.0	1356.0	851.6	261.0	404.3	137.8	28.0	4.7	48.2
2017	420.6	7975.0	1594.0	1708.0	681.2	372.0	128.8	181.9	77.5	2.0	31.7
2018	589.0	4030.0	4004.0	759.9	861.2	223.9	171.9	75.1	72.9	23.7	9.9
2019	582.4	14510.0	2186.0	1863.0	365.6	350.5	91.7	104.3	33.4	38.6	30.3
2020	520.4	2852.0	7625.0	1520.0	1055.0	194.4	230.4	91.6	60.0	46.6	68.5
2021	448.1	2900.0	1333.0	2593.0	587.4	388.1	84.6	73.0	10.4	8.5	46.4
2022	600.7	4828.0	2241.0	597.4	1153.0	277.8	165.2	31.9	29.6	8.6	24.7
2023	536.4	6431.0	2066.0	885.8	324.6	589.8	138.7	131.3	20.8	20.0	11.1

Year	0	1	2	3	4	5	6	7	8	9	10+
2024	1143.0	3536.0	2644.0	865.2	484.2	160.3	266.2	76.9	60.5	1.0	42.7

Table 16.8b. Sole in Subarea 4. Index of abundance, based on the BTS Q3 survey samples from The Netherlands, Germany, Belgium and the United Kingdom, used in the assessment of sole in Subarea 27.4.

Year	BTS Q3
1990	15776.37
1991	13543.50
1992	30152.34
1993	17377.76
1994	12505.74
1995	12988.56
1996	9237.29
1997	18001.27
1998	10138.25
1999	11646.50
2000	7903.99
2001	7871.93
2002	7563.86
2003	7543.57
2004	5077.29
2005	5449.56
2006	8107.10
2007	7355.46
2008	7600.85
2009	8117.42
2010	7428.02
2011	9905.25
2012	9671.46
2013	7565.44
2014	11710.30
2015	11510.99

Year	BTS Q3
2016	8367.18
2017	13172.48
2018	10821.62
2019	20150.65
2020	14264.07
2021	8473.09
2022	9957.70
2023	11155.60
2024	9280.10

Table 16.9a. Sole in Subarea 4. Index of abundance at age (COAST), based on the combined SNS and DYFS survey samples from The Netherlands, Germany, Belgium, used in the assessment of sole in Subarea 27.4.

Year	0	1	2	3	4	5
1990	2488	2040	1300	821.5	50.66	21.74
1991	16050	710.3	2428	473.5	559.1	12.93
1992	2479	8527	444.6	596.8	151.3	240.4
1993	2918	1677	4085	98.36	257	90.19
1994	3075	706.4	672.5	1097	5.686	64.34
1995	2080	799.9	386	223.9	300.9	7.367
1996	7999	511.1	527.7	94.13	62.23	82.17
1997	6119	7151	576.5	242.8	66.29	61.51
1998	3251	2010	2042	95.09	60.48	3.722
1999	5419	1097	1561	848.1	12.17	18.46
2000	3374	1460	529.2	230.9	103.5	6.904
2001	8428	862.9	1048	138.4	68.5	39.48
2002	4862	3100	597.2	344.9	59.07	20.58
2003	2553	2197	1480	210.2	135.2	13.46
2004	1820	734.4	524.4	377.4	56.09	49.14
2005	5856	507.2	297.6	113.1	46.57	0.4559
2006	1425	2595	222.2	20.43	47.29	29.23
2007	3493	638.6	1308	33.46	18.34	4.479

Year	0	1	2	3	4	5
2008	3411	617	472.3	435.5	13.69	0.5828
2009	4754	931.6	393.2	144.1	194	3.564
2010	6031	1811	571.4	102.5	37.37	41.34
2011	2273	1814	970.5	214.8	37.62	21.25
2012	2358	355.3	891.9	267.9	39.37	7.911
2013	3413	1030	476.2	578.9	209.7	31.14
2014	2466	1267	610.4	63.5	146.5	53.08
2015	2193	1006	1036	277.3	31.54	74.64
2016	2230	761.7	911.9	456.6	131.8	11.44
2017	3589	752.9	456.2	314.7	188.3	34.74
2018	6092	780.2	479.3	185.1	121.7	54.65
2019	1854	4019	671.9	240.2	57.65	34.32
2020	2329	537.4	3068	378.4	129.9	21.43
2021	2078	433.7	218.2	630.8	146.1	53.54
2022	3087	885.8	338.8	96.39	274.7	69.14
2023	1220	1259	491.1	96.19	21.6	71.91
2024	4255	583.7	553.7	118.8	35.36	7.962

Table 16.9b. Sole in Subarea 4. Index of abundance (COAST), based on the combined SNS and DYFS survey samples from The Netherlands, Germany, Belgium, used in the assessment of sole in Subarea 27.4.

Year	COAST
1990	6722.04
1991	20237.27
1992	12439.25
1993	9125.82
1994	5620.77
1995	3797.91
1996	9276.08
1997	14217.32
1998	7462.65
1999	8954.90

Year	COAST
2000	5705.08
2001	10586.03
2002	8984.18
2003	6588.67
2004	3561.07
2005	6821.17
2006	4339.00
2007	5495.93
2008	4950.48
2009	6420.07
2010	8594.46
2011	5331.35
2012	3920.68
2013	5739.21
2014	4606.06
2015	4618.68
2016	4503.50
2017	5335.97
2018	7712.80
2019	6877.06
2020	6463.71
2021	3560.05
2022	4751.59
2023	3159.70
2024	5554.33

Table 16.10. Sole in Subarea 4. Time series of abundances at age (stock numbers in thousands) estimated by the SS3 stock assessment.

Year	0	1	2	3	4	5	6	7	8	9	10+
1957	1901560	686611	218441	193377	137203	31061	29044	70170	23070	12147	25901
1958	8749880	517200	323037	135983	125417	86517	20102	19119	46703	15463	25760

Year	0	1	2	3	4	5	6	7	8	9	10+
1959	627009	2379860	243332	201477	88489	76598	54232	12817	12325	30317	27022
1960	1179620	170538	1119670	150077	130048	54857	48736	35097	8386	8121	38117
1961	508888	320841	80234	693891	93814	76940	33309	30100	21916	5274	29396
1962	431271	138411	150949	50035	438566	53655	45163	19888	18170	13323	21311
1963	10367200	117300	65119	93637	33844	250206	31417	26898	11976	11018	21206
1964	1582260	2819740	55186	40353	58304	18226	138288	17662	15289	6855	18629
1965	636735	430355	1326610	34510	26963	36019	11556	89185	11517	10039	16909
1966	1179240	173184	202456	797779	23013	16535	22670	7398	57728	7507	17738
1967	1539750	320739	81466	119400	442727	14532	10717	14945	4931	38747	17118
1968	778931	418792	150865	48435	70675	248744	8380	6286	8863	2945	33625
1969	2104170	211859	196411	81793	22623	38811	140197	4804	3643	5173	21593
1970	597263	572308	99377	104995	38486	12451	21923	80549	2791	2131	15828
1971	1135780	162448	268416	57835	51211	22259	7391	13237	49172	1716	11166
1972	1652450	308918	76138	143143	28377	27986	12485	4216	7635	28562	7567
1973	1731440	449445	145088	42301	69170	16046	16242	7370	2517	4589	21873
1974	582152	470929	211010	81732	19639	36516	8694	8951	4107	1412	15012
1975	1623910	158338	221456	121081	39008	10453	19948	4831	5029	2323	9399
1976	2232370	441682	74315	121882	60800	19715	5422	10525	2577	2701	6366
1977	712214	607176	207126	44095	62110	31309	10420	2915	5720	1411	5014
1978	325644	193713	284508	112807	21310	31978	16544	5600	1584	3130	3553
1979	2016120	88571	91080	159806	54104	10776	16596	8734	2989	851	3626
1980	2091110	548360	41635	51287	77502	26418	5400	8460	4501	1551	2349
1981	1980770	568754	257598	25157	26830	38505	13471	2801	4436	2377	2080
1982	1952690	538745	267248	138859	14755	13688	20162	7175	1508	2406	2438
1983	1008930	531108	251776	147808	58390	7547	7186	10767	3874	820	2657
1984	1273370	274417	249571	133018	70646	26576	3526	3415	5172	1874	1700
1985	2351380	346339	128981	133870	58329	31593	12198	1646	1612	2459	1714
1986	820464	639546	162813	68824	59270	26229	14580	5726	781	770	2011
1987	6968890	223156	300671	95106	33942	26184	11892	6724	2670	367	1320

Year	0	1	2	3	4	5	6	7	8	9	10+
1988	1710140	1895450	104941	166878	48081	16966	13433	6206	3548	1419	906
1989	2584520	465136	891679	56350	78125	22664	8208	6610	3087	1777	1174
1990	1012710	702956	218770	525748	25748	38938	11593	4271	3477	1636	1577
1991	6468780	275443	330240	129385	278902	12263	19033	5764	2147	1760	1641
1992	1201880	1759430	129499	198169	75879	131040	5913	9335	2858	1072	1714
1993	1161730	326896	827097	75744	103775	37056	65680	3015	4812	1484	1460
1994	1477920	315976	153680	476768	38141	47686	17476	31507	1462	2350	1451
1995	784281	401976	148222	90908	252155	17478	22428	8360	15239	712	1867
1996	3641070	213314	186418	82516	54704	119210	8481	11069	4172	7658	1309
1997	1575050	990325	100197	103949	43378	26680	59672	4318	5698	2163	4681
1998	1329340	428393	465048	58946	54134	23309	14714	33473	2449	3254	3948
1999	1754270	361564	201357	254748	29199	28407	12554	8061	18540	1366	4055
2000	1118820	477139	169930	116458	122871	15499	15476	6956	4516	10460	3091
2001	2699410	304304	223286	96828	59381	57776	7480	7597	3452	2257	6823
2002	1456500	734205	142677	124448	52114	28880	28839	3798	3900	1785	4743
2003	867692	396149	331914	70178	54746	25037	14852	15727	2094	2165	3662
2004	899112	236001	183437	178517	36346	27790	13234	8100	8673	1163	3268
2005	2571760	244547	107601	92099	80100	17458	14154	7082	4382	4725	2439
2006	935166	699487	111906	56971	43596	35087	8033	6778	3429	2137	3522
2007	1292340	254353	319974	58311	28029	20900	17611	4183	3569	1818	3029
2008	1396780	351499	118071	162532	30664	14320	11093	9622	2311	1985	2723
2009	2209230	379906	162175	66395	80949	16178	7822	6217	5452	1319	2712
2010	2663710	600882	175555	87053	35934	38673	8023	3991	3207	2832	2115
2011	760145	724495	275975	90453	43567	16633	18846	4078	2051	1660	2582
2012	1173710	206750	336975	156883	43426	21518	8538	9963	2180	1104	2306
2013	1920610	319235	96075	196606	83412	20478	10557	4319	5096	1123	1774
2014	1392820	522381	147919	54848	110917	38935	9976	5319	2200	2614	1500
2015	888837	378830	241937	81544	32403	54511	19954	5283	2848	1186	2237
2016	1670380	241752	175806	135283	42613	16070	28139	10625	2844	1544	1874

Year	1	2	3	4	5	6	7	8	9	10+
1976	0.003	0.057	0.306	0.343	0.343	0.343	0.343	0.343	0.343	0.343
1977	0.004	0.143	0.359	0.343	0.343	0.343	0.343	0.343	0.343	0.343
1978	0.001	0.112	0.367	0.361	0.361	0.361	0.361	0.361	0.361	0.361
1979	0.001	0.109	0.356	0.396	0.396	0.396	0.396	0.396	0.396	0.396
1980	0.002	0.039	0.280	0.379	0.379	0.379	0.379	0.379	0.379	0.379
1981	0.001	0.153	0.166	0.352	0.352	0.352	0.352	0.352	0.352	0.352
1982	0.007	0.127	0.498	0.349	0.349	0.349	0.349	0.349	0.349	0.349
1983	0.001	0.173	0.370	0.466	0.466	0.466	0.466	0.466	0.466	0.466
1984	0.001	0.158	0.456	0.484	0.484	0.484	0.484	0.484	0.484	0.484
1985	0.001	0.163	0.447	0.478	0.478	0.478	0.478	0.478	0.478	0.478
1986	0.001	0.073	0.339	0.496	0.496	0.496	0.496	0.496	0.496	0.496
1987	0.000	0.124	0.314	0.372	0.372	0.372	0.372	0.372	0.372	0.372
1988	0.000	0.157	0.391	0.431	0.431	0.431	0.431	0.431	0.431	0.431
1989	0.000	0.063	0.415	0.375	0.375	0.375	0.375	0.375	0.375	0.375
1990	0.001	0.060	0.266	0.421	0.421	0.421	0.421	0.421	0.421	0.421
1991	0.001	0.046	0.166	0.434	0.434	0.434	0.434	0.434	0.434	0.434
1992	0.001	0.071	0.279	0.396	0.396	0.396	0.396	0.396	0.396	0.396
1993	0.001	0.086	0.318	0.457	0.457	0.457	0.457	0.457	0.457	0.457
1994	0.003	0.060	0.269	0.459	0.459	0.459	0.459	0.459	0.459	0.459
1995	0.014	0.121	0.140	0.428	0.428	0.428	0.428	0.428	0.428	0.428
1996	0.002	0.119	0.275	0.397	0.397	0.397	0.397	0.397	0.397	0.397
1997	0.002	0.066	0.284	0.300	0.300	0.300	0.300	0.300	0.300	0.300
1998	0.001	0.137	0.334	0.324	0.324	0.324	0.324	0.324	0.324	0.324
1999	0.001	0.083	0.361	0.312	0.312	0.312	0.312	0.312	0.312	0.312
2000	0.005	0.097	0.306	0.434	0.434	0.434	0.434	0.434	0.434	0.434
2001	0.003	0.120	0.251	0.400	0.400	0.400	0.400	0.400	0.400	0.400
2002	0.040	0.245	0.453	0.412	0.370	0.328	0.328	0.328	0.328	0.328
2003	0.016	0.155	0.290	0.357	0.343	0.328	0.328	0.328	0.328	0.328
2004	0.031	0.224	0.433	0.412	0.380	0.347	0.347	0.347	0.347	0.347

Year	1	2	3	4	5	6	7	8	9	10+
2005	0.028	0.171	0.380	0.504	0.481	0.458	0.458	0.458	0.458	0.458
2006	0.028	0.187	0.341	0.414	0.394	0.375	0.375	0.375	0.375	0.375
2007	0.013	0.212	0.275	0.351	0.338	0.326	0.326	0.326	0.326	0.326
2008	0.020	0.111	0.329	0.318	0.310	0.301	0.301	0.301	0.301	0.301
2009	0.018	0.157	0.246	0.418	0.406	0.395	0.395	0.395	0.395	0.395
2010	0.024	0.198	0.324	0.449	0.424	0.399	0.399	0.399	0.399	0.399
2011	0.011	0.100	0.366	0.384	0.372	0.359	0.359	0.359	0.359	0.359
2012	0.012	0.074	0.264	0.431	0.417	0.404	0.404	0.404	0.404	0.404
2013	0.015	0.096	0.204	0.441	0.424	0.408	0.408	0.408	0.408	0.408
2014	0.016	0.131	0.158	0.389	0.373	0.358	0.358	0.358	0.358	0.358
2015	0.014	0.116	0.281	0.380	0.366	0.352	0.352	0.352	0.352	0.352
2016	0.013	0.098	0.283	0.481	0.467	0.453	0.453	0.453	0.453	0.453
2017	0.017	0.128	0.250	0.422	0.404	0.387	0.387	0.387	0.387	0.387
2018	0.016	0.131	0.237	0.419	0.403	0.387	0.387	0.387	0.387	0.387
2019	0.016	0.109	0.217	0.372	0.355	0.338	0.338	0.338	0.338	0.338
2020	0.010	0.081	0.144	0.272	0.263	0.253	0.253	0.253	0.253	0.253
2021	0.010	0.066	0.140	0.189	0.179	0.170	0.170	0.170	0.170	0.170
2022	0.008	0.058	0.049	0.122	0.114	0.107	0.107	0.107	0.107	0.107
2023	0.008	0.053	0.072	0.066	0.062	0.057	0.057	0.057	0.057	0.057
2024	0.005	0.052	0.089	0.054	0.050	0.046	0.046	0.046	0.046	0.046

Table 16.12. Sole in Subarea 4. Catch scenarios. All weights are in tonnes.

Basis	Total catch* (2026)	Projected landings (2026)	Projected discards** (2026)	F _{total} *** (ages 2–6) (2026)	F _{projected landings} (ages 2–6) (2026)	F _{projected discards} (ages 1–3) (2026)	Spawning-stock biomass (SSB; 2027)	% SSB change^	% TAC change^^	% advice change^^^
ICES advice basis										
MSY approach: F _{MSY}	12454	11060	1394	0.186	0.157	0.031	56850	7.0	25	16.4
Other scenarios										
F _{MSY upper}	12454	11060	1394	0.186	0.157	0.031	56850	7.0	25	16.4
F _{MSY lower}	11340	10074	1266	0.168	0.142	0.028	57774	8.8	13.4	6.0
F = 0	0	0	0	0	0	0	67241	27	–100	–100
F _{pa}	12454	11060	1394	0.186	0.157	0.031	56850	7.0	25	16.4
SSB (2027) = B _{pa} = MSY B _{trigger}	22792	20185	2607	0.37	0.31	0.061	48340	–9.0	128	113
SSB (2027) = B _{lim}	39636	34901	4735	0.75	0.63	0.125	34788	–35	296	271
F = F ₂₀₂₅	5426	4827	599	0.077	0.065	0.0128	62697	18.0	–46	–49
Rollover advice	10696	9503	1193	0.158	0.133	0.026	58308	9.8	7.0	0

* Any difference between the total catch and the sum of projected landings and discards result from rounding.

** Including below minimum size (BMS) landings. Assuming average estimated discard rate by age 2022–2024.

*** F_{projected landings} and F_{projected discards} do not add up to F_{total}, as they are calculated using different ages.

^ SSB 2027 relative to SSB 2026.

^^ Total advised catch in 2026 relative to the TAC in 2025 (10 000 tonnes).

^^^ Total advised catch in 2026 relative to the advice value 2025 (10 696 tonnes).

Table 16.13. Sole in Subarea 4. Assessment summary. Recruitment is in thousands; all weights are in tonnes. “Low” and “High” refer to 2 standard deviations.

Year	Recruitment (age 0)			Spawning-stock biomass (SSB)			Land-ings	Dis-cards*	Fishing pressure (ages 2–6)		
	Low	R	High	Low	SSB	High			Low	F	High
1957	1084199	1901560	2718921	93277	108658	124039	12067		0.084	0.099	0.114
1958	7382578	8749880	10117182	93720	109625	125530	14287		0.099	0.117	0.135
1959	230327	627009	1023691	94486	110401	126316	13832		0.095	0.112	0.129
1960	736670	1179620	1622570	90884	106703	122522	18620		0.122	0.145	0.169
1961	206370	508888	811406	155238	174754	194270	23566		0.130	0.162	0.194
1962	154552	431271	707990	124266	142068	159870	26877		0.127	0.151	0.176
1963	9137257	10367200	11597143	98511	116029	133547	26164		0.167	0.20	0.24
1964	885866	1582260	2278654	69032	84906	100781	11342		0.082	0.104	0.127
1965	318572	636735	954898	57728	72555	87383	17043		0.087	0.117	0.147
1966	800907	1179240	1557573	163364	181148	198932	33340		0.098	0.140	0.182
1967	1129667	1539750	1949833	145497	160841	176185	33439		0.170	0.196	0.22
1968	457587	778931	1100275	116006	130954	145902	33179		0.21	0.28	0.34
1969	1683319	2104170	2525021	87995	101509	115023	27559		0.23	0.28	0.32
1970	335427	597263	859099	76473	88930	101387	19684		0.189	0.22	0.25
1971	796635	1135780	1474925	63526	75220	86913	23652		0.22	0.27	0.32
1972	1219163	1652450	2085737	70111	81658	93206	21087		0.21	0.25	0.29
1973	1307253	1731440	2155627	52864	63108	73352	19308		0.23	0.29	0.35
1974	311939	582152	852365	49381	59736	70092	17990		0.22	0.28	0.33
1975	1219756	1623910	2028064	52650	63021	73392	20773		0.24	0.31	0.37
1976	1764837	2232370	2699903	55609	65406	75203	17326		0.22	0.28	0.33
1977	427208	712214	997220	41996	50828	59660	18003		0.25	0.31	0.37
1978	133641	325644	517647	46375	55738	65101	20280		0.24	0.31	0.38
1979	1557666	2016120	2474574	59671	70270	80869	22598		0.26	0.33	0.40
1980	1559150	2091110	2623070	44218	54027	63835	15806		0.23	0.29	0.35
1981	1491991	1980770	2469549	29827	38860	47893	15403		0.20	0.27	0.35
1982	1504344	1952690	2401036	41214	50748	60281	21578		0.24	0.33	0.43

Year	Recruitment (age 0)			Spawning-stock biomass (SSB)			Land-ings	Dis-cards*	Fishing pressure (ages 2–6)		
	Low	R	High	Low	SSB	High			Low	F	High
1983	640962	1008930	1376898	51232	60787	70342	24927		0.29	0.39	0.49
1984	845478	1273370	1701262	54380	63458	72537	26839		0.33	0.41	0.49
1985	1831806	2351380	2870954	50775	59320	67866	24248		0.33	0.41	0.49
1986	462270	820464	1178658	41002	49886	58769	18201		0.30	0.38	0.46
1987	6230613	6968890	7707167	36730	45819	54907	17368		0.23	0.31	0.39
1988	1284014	1710140	2136266	51963	61438	70912	21590		0.28	0.37	0.46
1989	2221808	2584520	2947232	40763	49993	59222	21804		0.25	0.32	0.40
1990	809280	1012710	1216140	126737	138618	150499	35121		0.21	0.32	0.42
1991	6021994	6468780	6915566	101921	111801	121681	33514		0.27	0.30	0.34
1992	988706	1201880	1415054	98188	107129	116070	29341		0.27	0.31	0.34
1993	984475	1161730	1338985	64784	71621	78459	31491		0.30	0.35	0.41
1994	1299737	1477920	1656103	114452	122292	130132	33002		0.27	0.34	0.41
1995	640976	784281	927586	91051	98309	105566	30468		0.28	0.31	0.34
1996	3374622	3641070	3907518	63664	69914	76164	22650		0.28	0.32	0.35
1997	1378272	1575050	1771828	54573	60307	66041	14902		0.22	0.25	0.28
1998	1164558	1329340	1494122	42680	47761	52842	20867		0.24	0.29	0.34
1999	1578971	1754270	1929569	70204	76114	82024	23475		0.21	0.28	0.34
2000	981625	1118820	1256015	60960	66431	71903	22641		0.30	0.34	0.38
2001	2499839	2699410	2898981	51812	57215	62618	19944		0.27	0.31	0.35
2002	1311335	1456500	1601665	51733	56783	61834	15944	3457	0.31	0.36	0.41
2003	766736	867692	968648	39380	43818	48256	15654	1902	0.25	0.29	0.34
2004	798553	899112	999671	50358	54883	59408	17932	3281	0.30	0.36	0.42
2005	2398475	2571760	2745045	41928	45826	49724	15777	1885	0.35	0.40	0.45
2006	826950	935166	1043382	34155	38064	41974	12216	1986	0.30	0.34	0.39
2007	1166011	1292340	1418669	24341	27286	30230	14439	1406	0.25	0.30	0.35
2008	1259681	1396780	1533879	46967	50709	54451	14206	897	0.22	0.27	0.33
2009	2024916	2209230	2393544	39898	43308	46718	14282	1210	0.29	0.32	0.36
2010	2454754	2663710	2872666	32837	35839	38841	12793	2896	0.31	0.36	0.41

Year	Recruitment (age 0)			Spawning-stock biomass (SSB)			Land-ings	Dis-cards*	Fishing pressure (ages 2–6)		
	Low	R	High	Low	SSB	High			Low	F	High
2011	654097	760145	866193	34645	37897	41149	11497	1863	0.27	0.32	0.36
2012	1047524	1173710	1299896	39732	42986	46239	12210	2854	0.26	0.32	0.37
2013	1757269	1920610	2083951	46386	49701	53016	13145	2336	0.27	0.31	0.36
2014	1260311	1392820	1525329	39089	42153	45217	13091	1949	0.25	0.28	0.31
2015	782175	888837	995499	34067	36966	39866	12920	1755	0.26	0.30	0.33
2016	1508913	1670380	1831847	39659	42706	45754	14133	1237	0.31	0.36	0.41
2017	1207094	1357460	1507826	36706	39734	42761	12373	1325	0.28	0.32	0.36
2018	3453790	3742770	4031750	28411	31182	33952	11193	1184	0.28	0.32	0.36
2019	579569	686690	793811	28319	31120	33920	8647	1963	0.23	0.28	0.32
2020	835423	952005	1068587	31802	35126	38450	8864	1472	0.168	0.20	0.24
2021	919570	1047720	1175870	56244	61186	66127	8238	955	0.122	0.149	0.176
2022	1362816	1535320	1707824	46199	50538	54876	5291	606	0.079	0.090	0.101
2023	727446	870509	1013572	49265	53803	58341	4029	386	0.055	0.062	0.069
2024	1647210	1993650	2340090	56943	61914	66886	3960	395	0.052	0.058	0.064
2025		1425136**			57728						

* Since 2016, discards include below minimum size (BMS) landings from EU and UK fleets.

** Geometric mean 1957–2023.

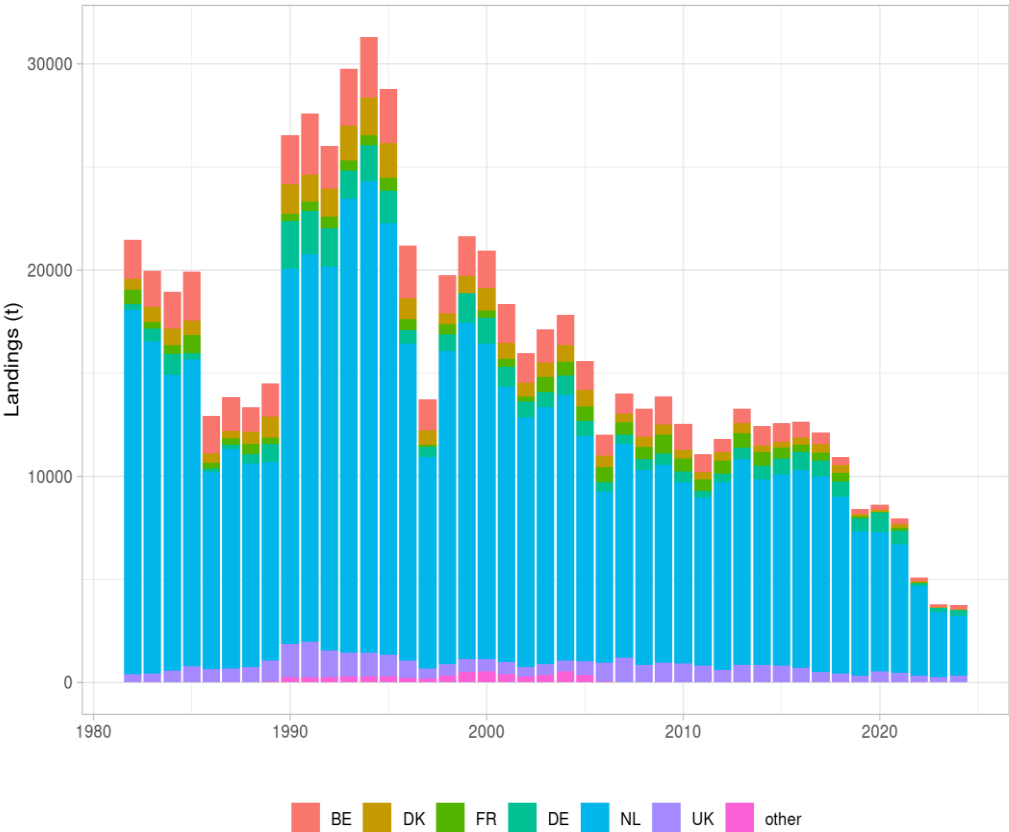


Figure 16.1. Sole in Subarea 4. Official landings reported to ICES by country.

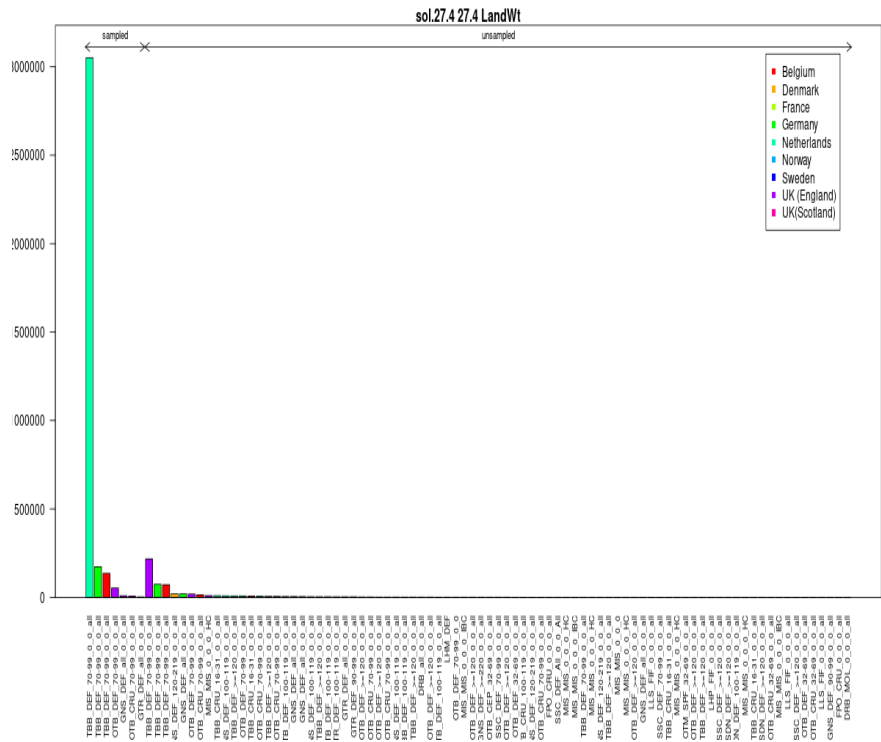


Figure 16.2. Sole in Subarea 4. InterCatch summary plots. Sampled and unsampled fleets for landings yield estimation (tonnes).



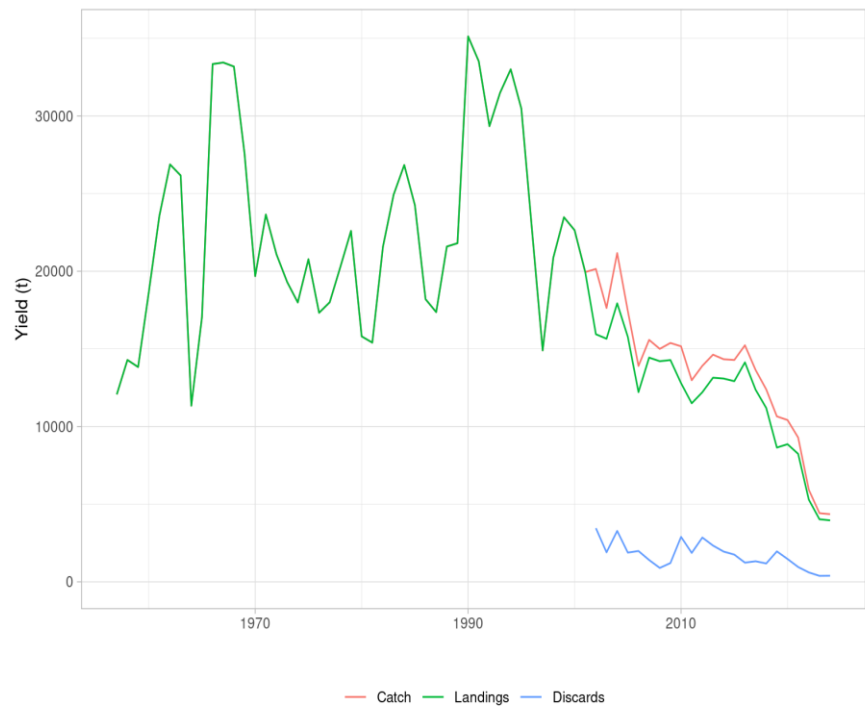


Figure 16.5. Sole in Subarea 4. Time-series of ICES catches, landings and discards (in tonnes).

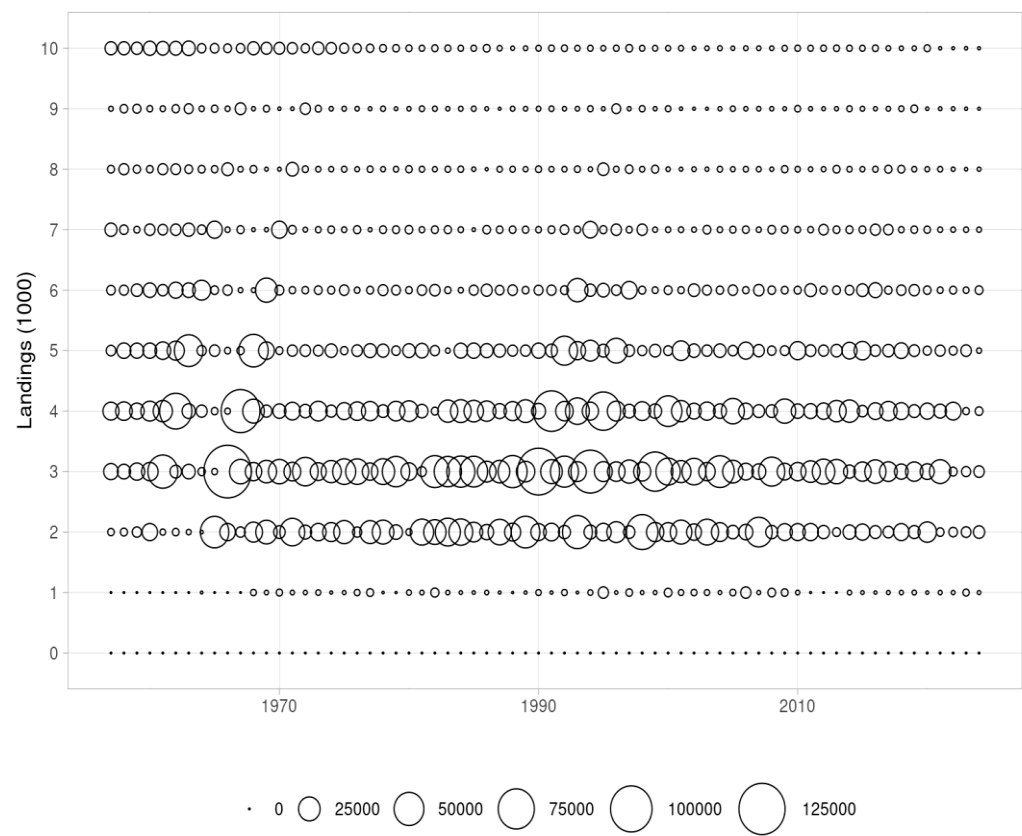


Figure 16.6. Sole in Subarea 4. Time series of landings at age (in thousands).

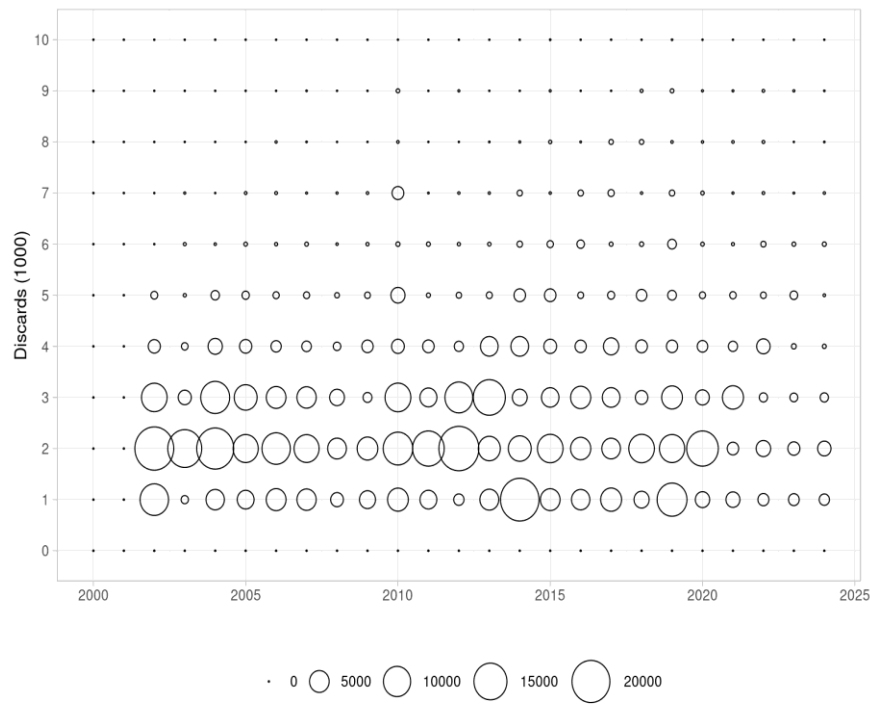


Figure 16.7. Sole in Subarea 4. Time-series of discards at age (in thousands).

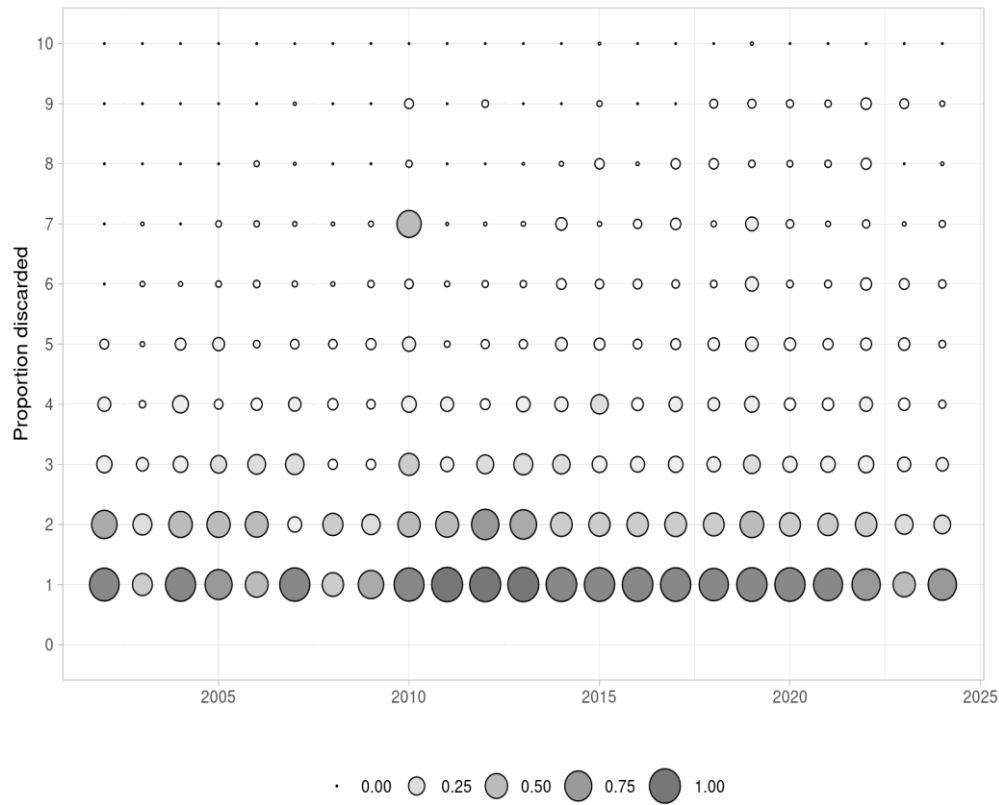


Figure 16.8. Sole in Subarea 4. Proportions of discarded sole by age over the period for which discards are available.

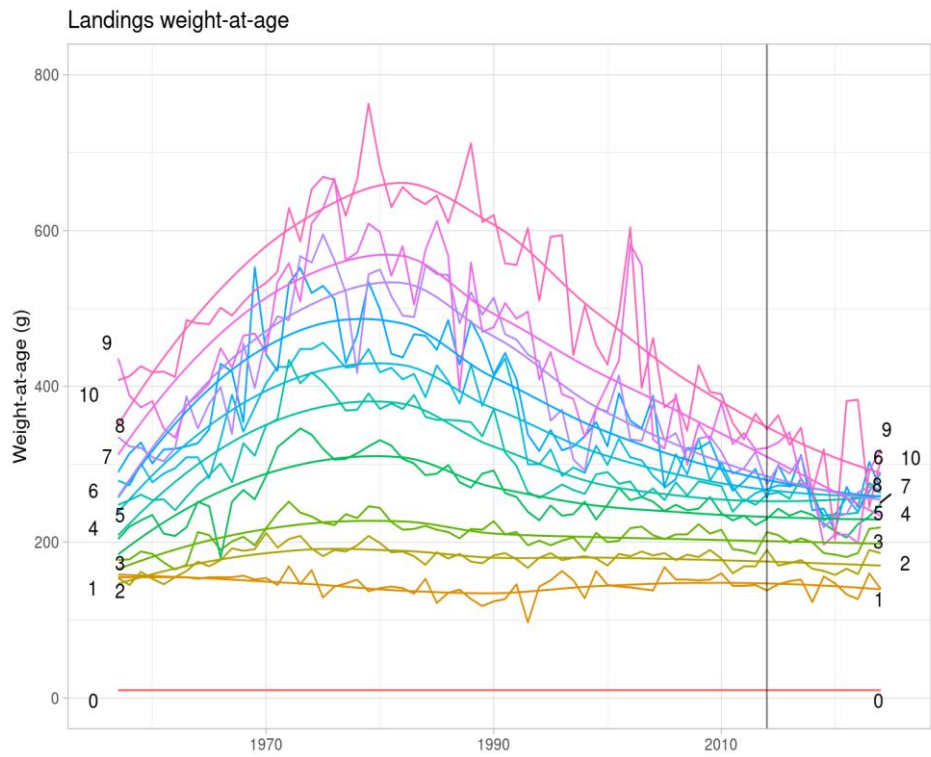


Figure 16.9. Sole in Subarea 4. Time-series of mean weight-at-age in the landings (in grammes).

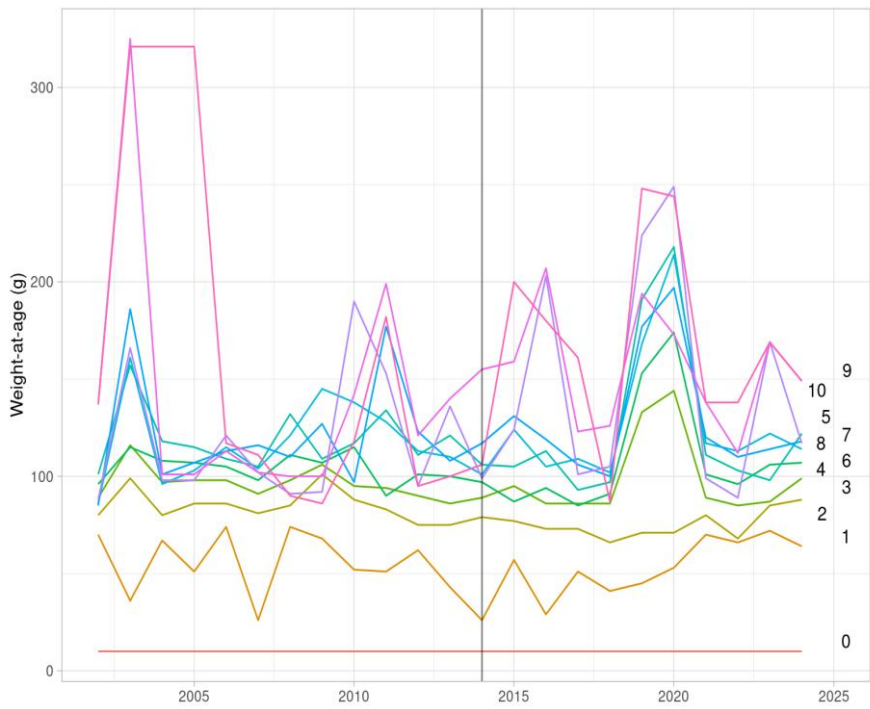


Figure 16.10. Sole in Subarea 4. Time-series of mean weight-at-age in the discards (in grammes).

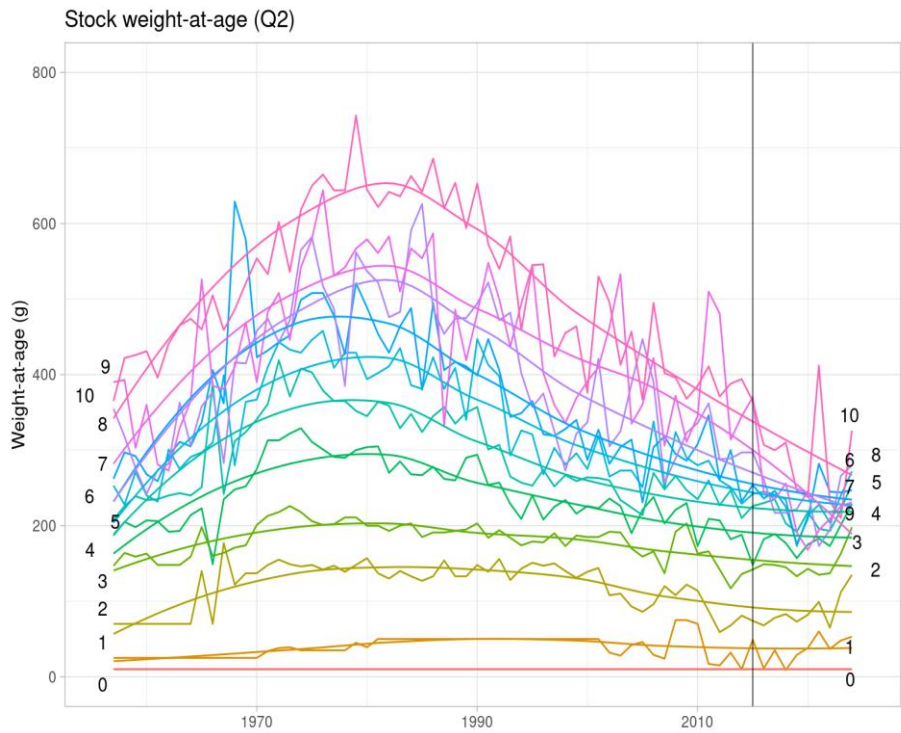


Figure 16.11. Sole in Subarea 4. Time-series of mean weight-at-age in the stock (in grammes).

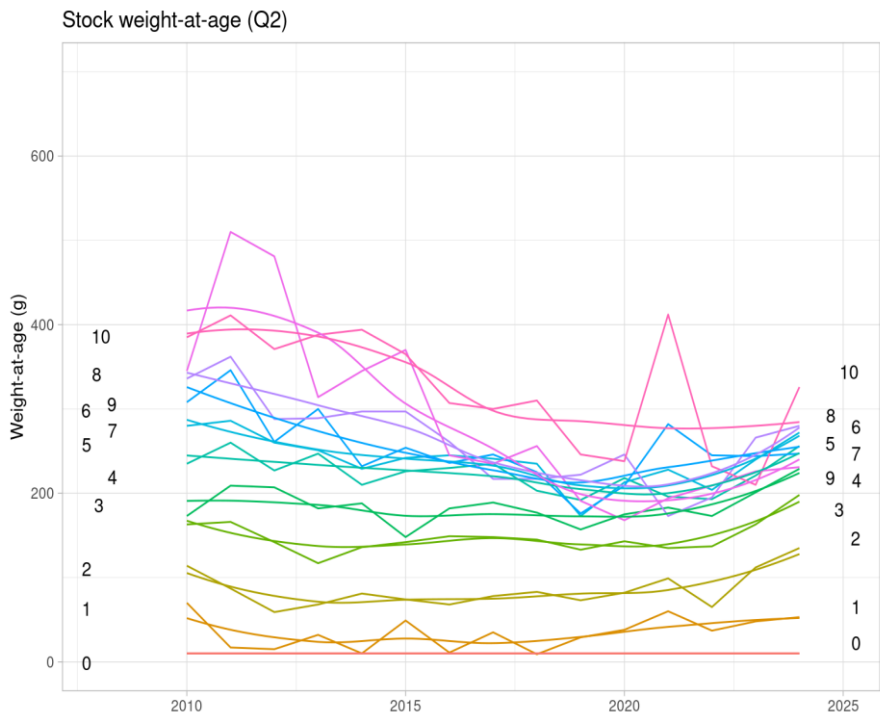


Figure 16.12. Sole in Subarea 4. Recent values of the time-series (2009–2024) of mean weight-at-age in the stock (in grammes).



Figure 16.13. Sole in Subarea 4. Recent values of the time series (2010-2024) of mean weight-at-age in the landings, discards and stock (in grammes).

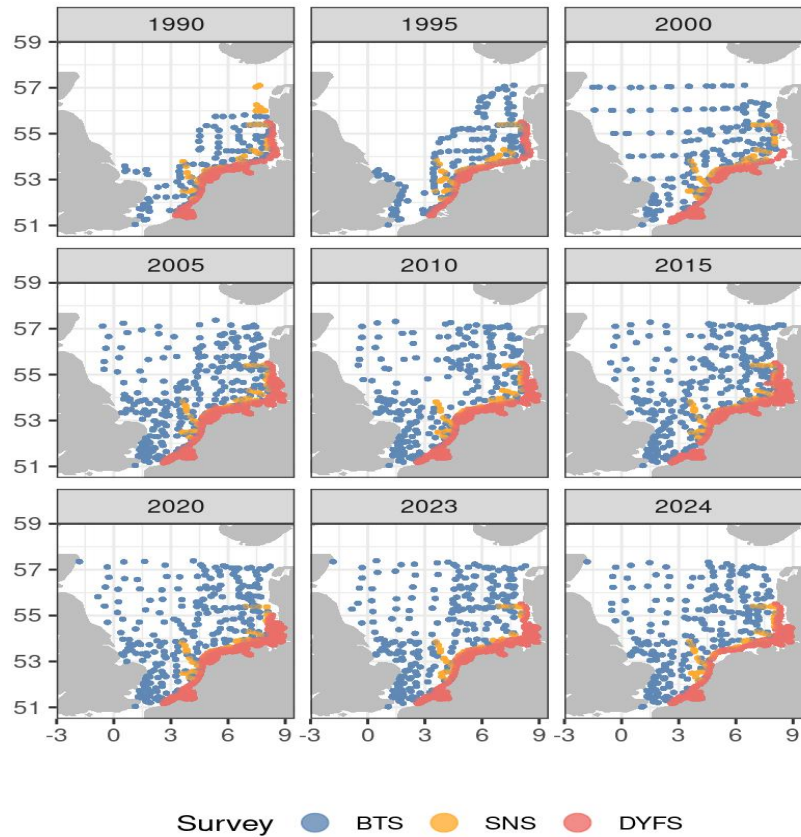


Figure 16.14. Sole in Subarea 4. Location of stations sampled during the BTS Q3 survey included in the BTS index of abundance and the SNS Q3 survey and DYFS Q3 survey, both included in the COAST index of abundance. Data shown for a selection of years.



Figure 16.15. Sole in Subarea 4. Comparison of the time series of the two indices of relative stock biomass: BTS Q3 delta-lognormal GAM standardized (1990–2024) and COAST (SNS + DYFS, 1990–2024).

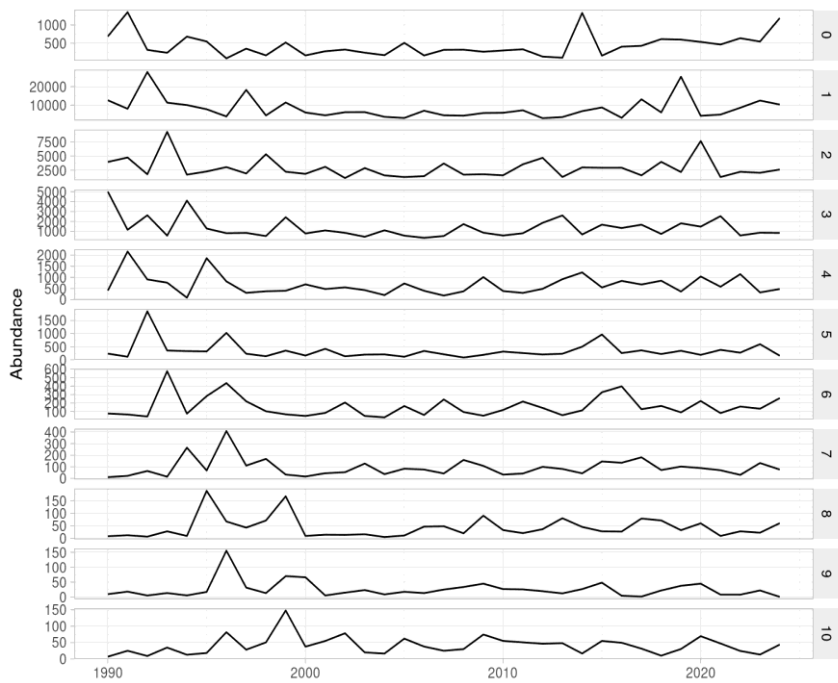


Figure 16.16. Sole in Subarea 4. Time series of relative abundance at age from the BTS Q3 delta-lognormal GAM standardized index of abundance (1990–2024).

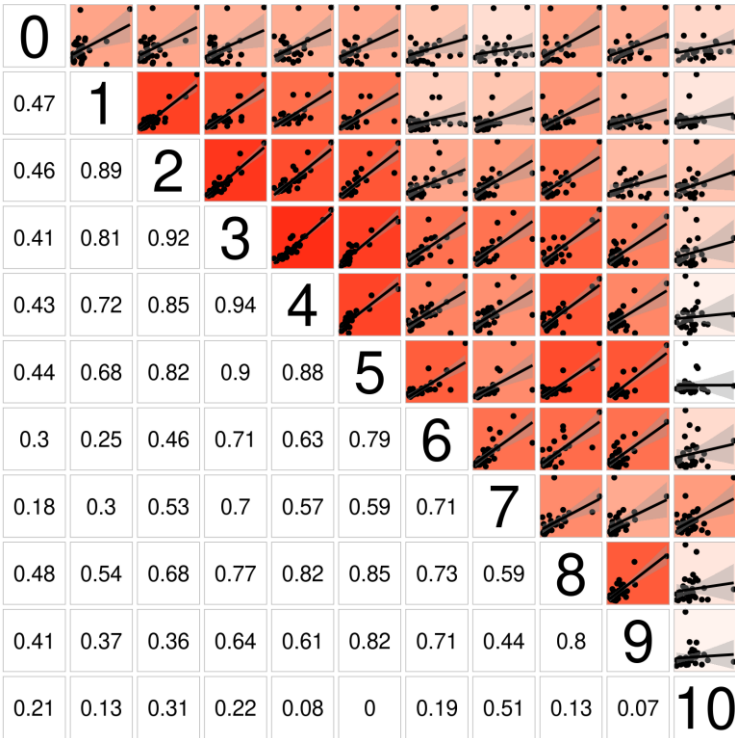


Figure 16.17. Sole in Subarea 4. Bivariate cross-correlation plots showing the internal consistency in signals by cohort for the BTS Q3 delta-lognormal GAM standardized index of abundance (1990–2024).

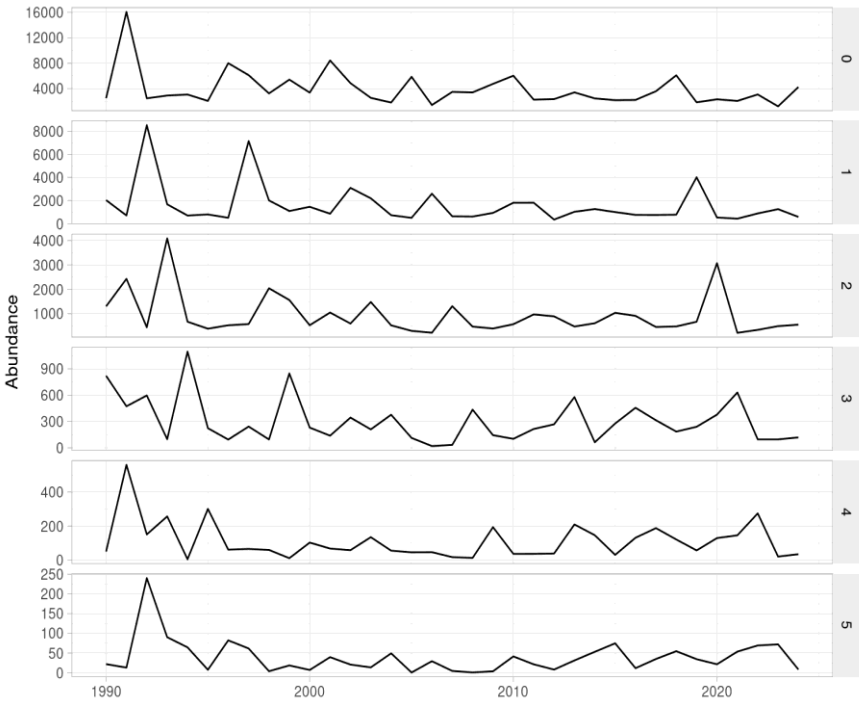


Figure 16.18. Sole in Subarea 4. Time series of relative abundance at age from the COAST (SNS + DYFS) index of abundance (1990–2024).

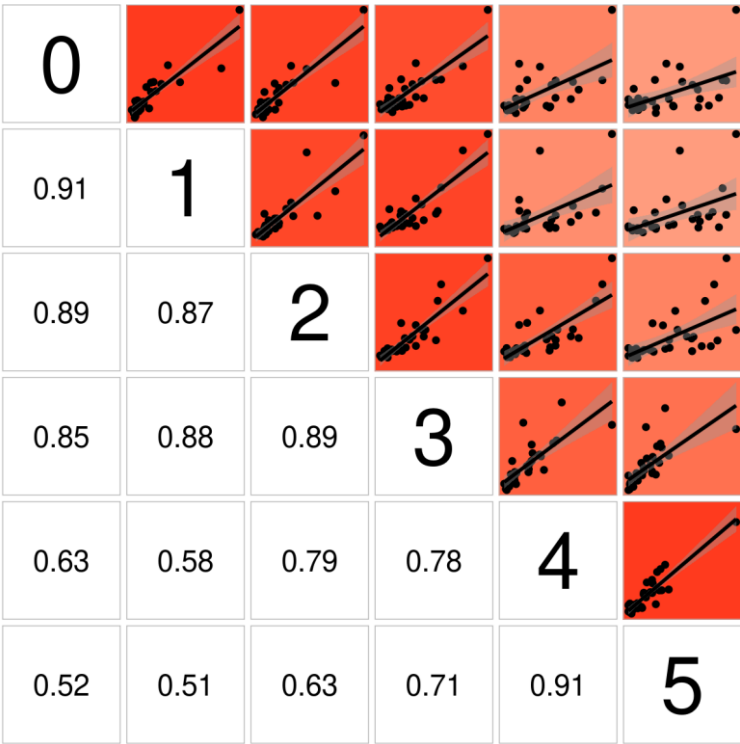


Figure 16.19. Sole in Subarea 4. Bivariate cross-correlation plots showing the internal consistency in signals by cohort for the COAST (SNS + DYFS) index of abundance (1990–2024).

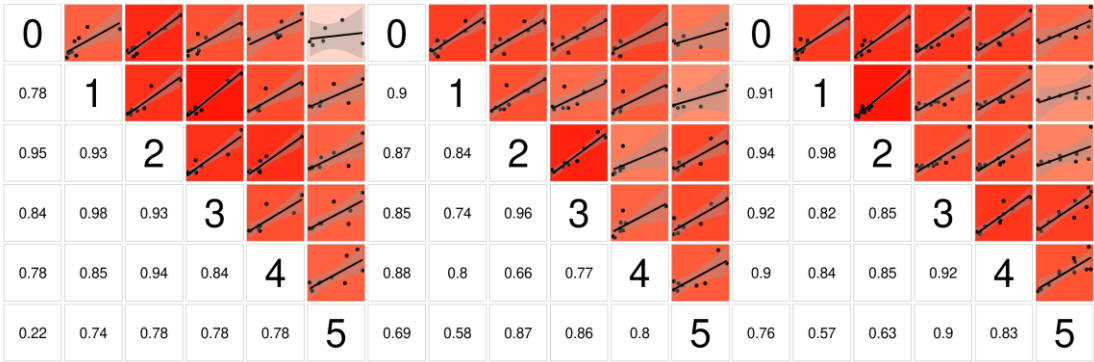


Figure 16.20. Sole in Subarea 4. Bivariate cross-correlation plots showing the internal consistency in signals by cohort for the COAST (SNS + DYFS) index of abundance, over the last three decades (left to right 1992–2002, 2003–2012 and 2013–2024).

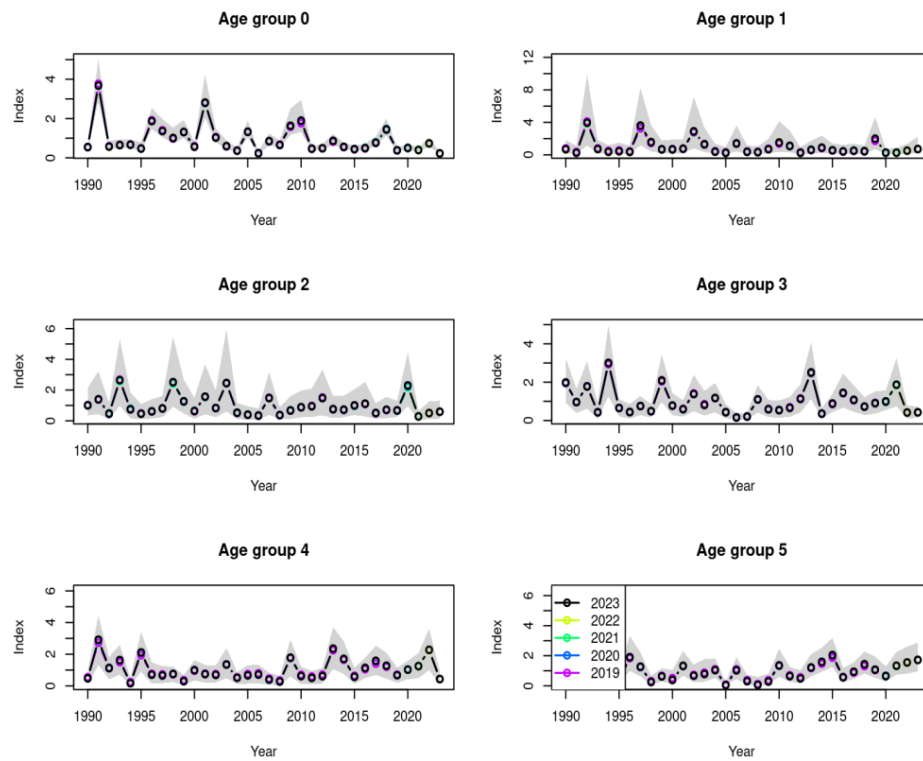


Figure 16.21. Sole in Subarea 4. Retrospective pattern in the lognormal GAM-standardized BTS Q3 numbers-at-age index of relative abundance.

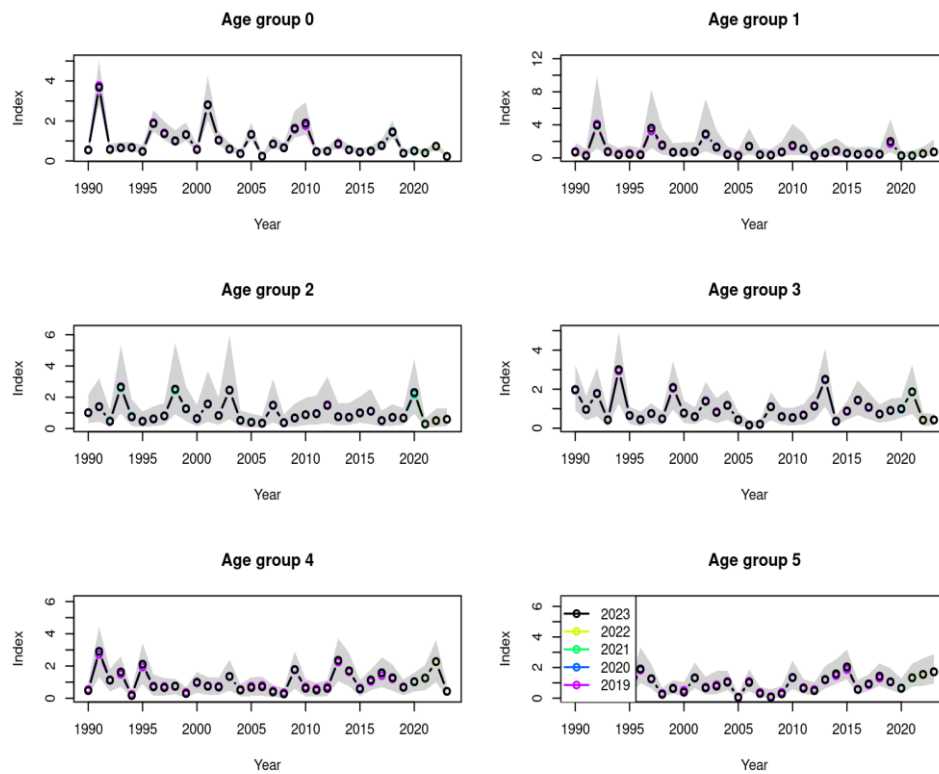


Figure 16.22. Sole in Subarea 4. Retrospective pattern in the lognormal GAM-standardized COAST (SNS + DYFS) numbers-at-age index of relative abundance.

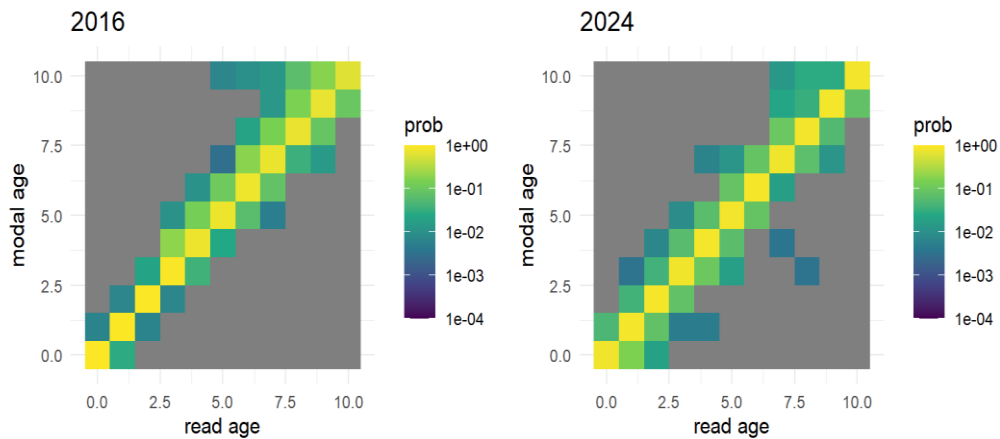


Figure 16.23. Sole in Subarea 4. Matrices of aging error for North sea sole obtained from the 2016 (ICES, 2016) and 2024 (ICES WGBIOP) otolith exchange exercises.

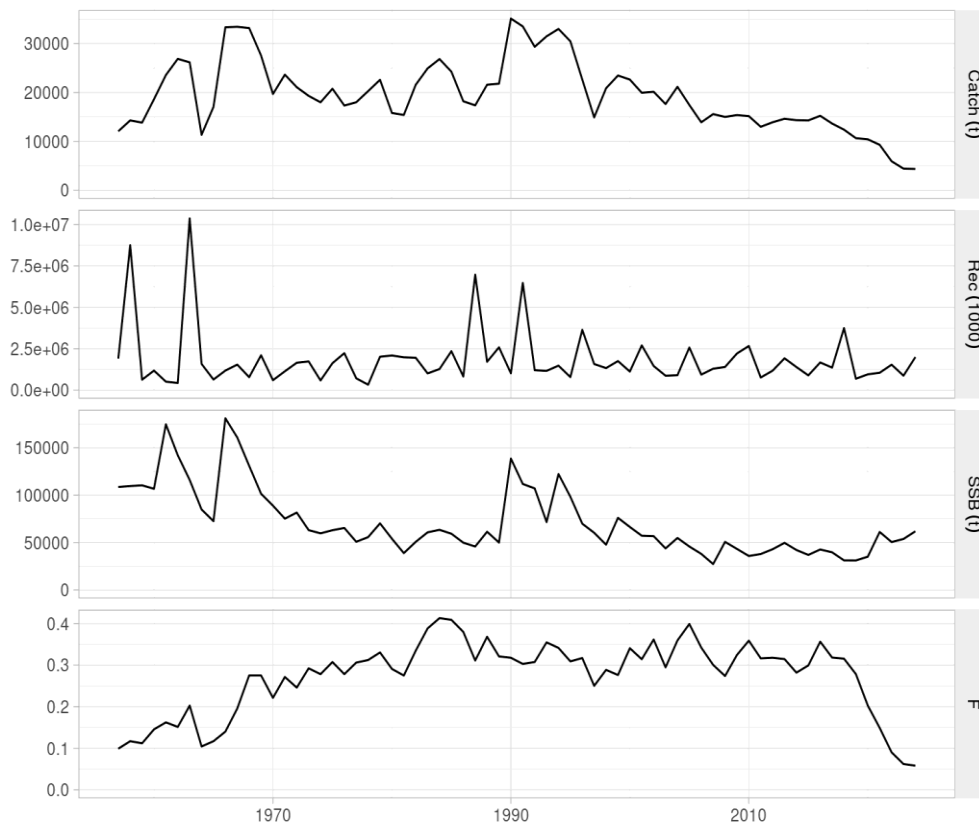


Figure 16.24. Sole in Subarea 4. Time series of recruitment at age 0 (in thousands), spawning stock biomass (in tonnes) and fishing mortality (as average of ages 2 to 6), together with total catch (in tonnes).

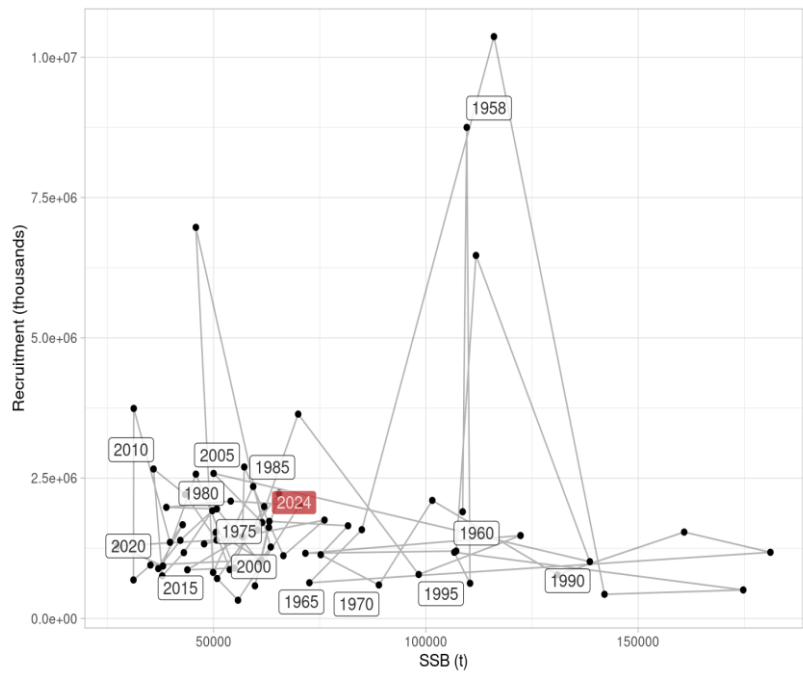


Figure 16.25. Sole in Subarea 4. Estimates of recruitment at age 0 (in thousands) and spawning biomass (in tonnes), connected in time. Labels refer to the year in which recruitment was observed.

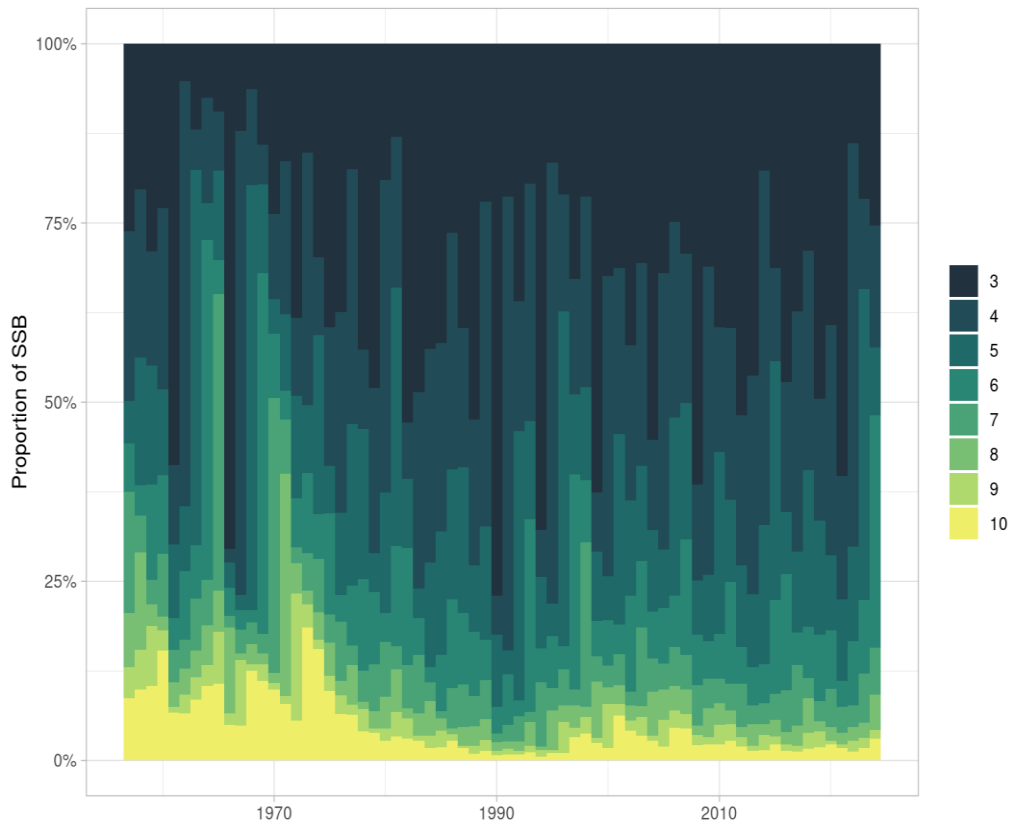


Figure 16.26. Sole in Subarea 4. Estimated proportions of spawning stock biomass by age and year.

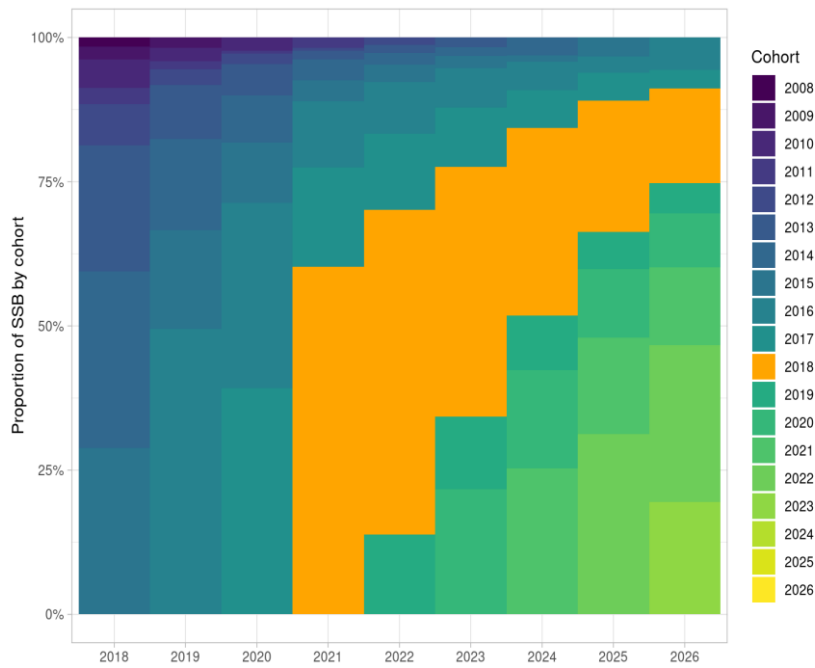


Figure 16.27. Sole in Subarea 4. Estimated proportions of spawning biomass by cohort and year.

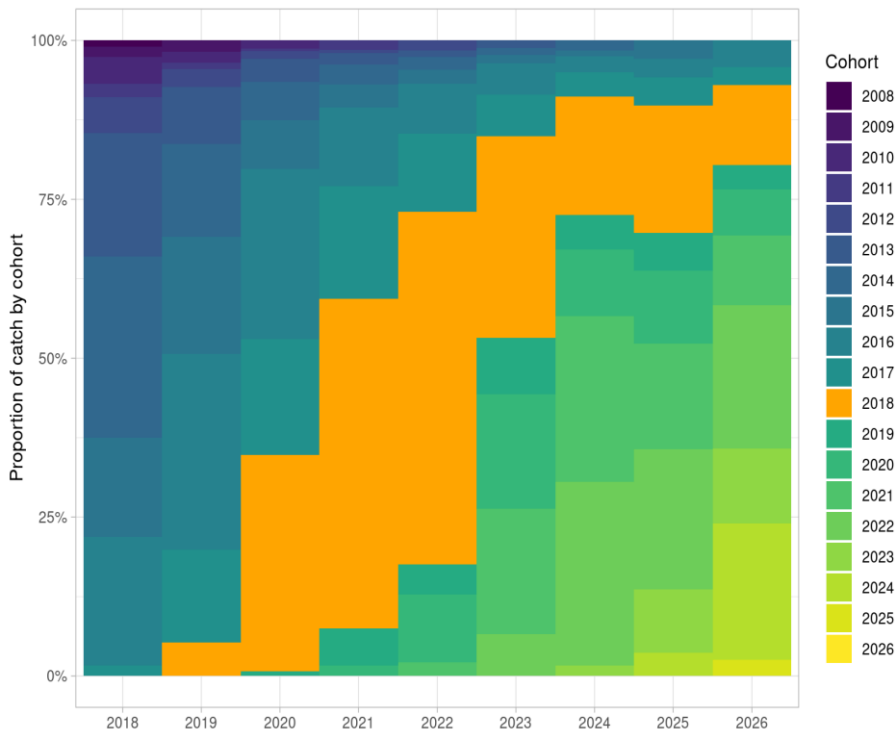


Figure 16.28. Sole in Subarea 4. Proportions in the catch by cohort and year.

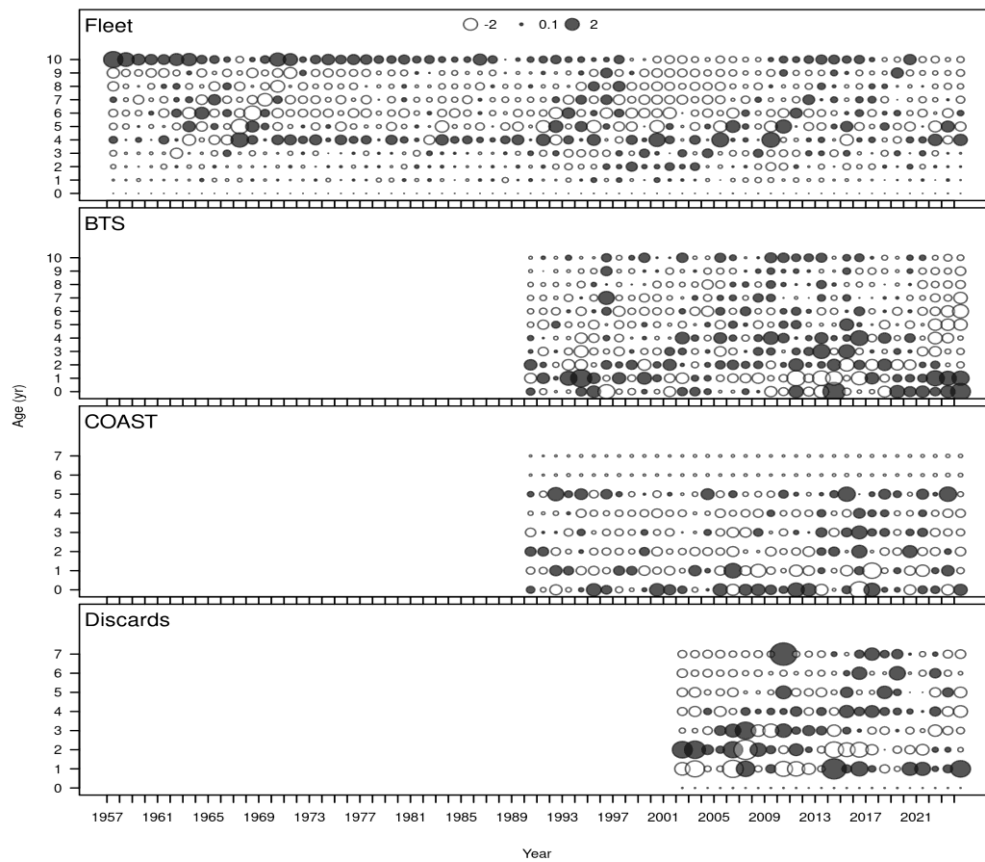


Figure 16.29. Sole in Subarea 4. Residuals of model fit to the four sources of data: BTS and COAST indices of abundance, landings-at-age (Fleet) and discards-at-age (Discards).

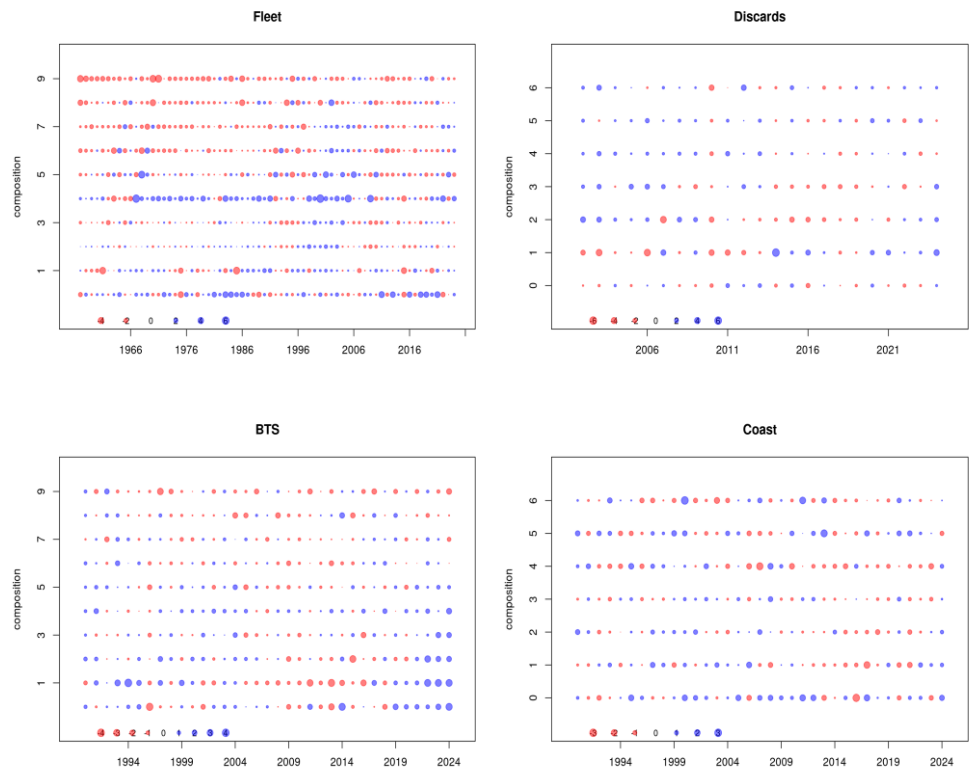


Figure 16.30. Sole in Subarea 4. One step ahead (OSA) residuals of model fit to the four sources of data: BTS and COAST indices of abundance, landings-at-age (Fleet) and discards-at-age (Discards).

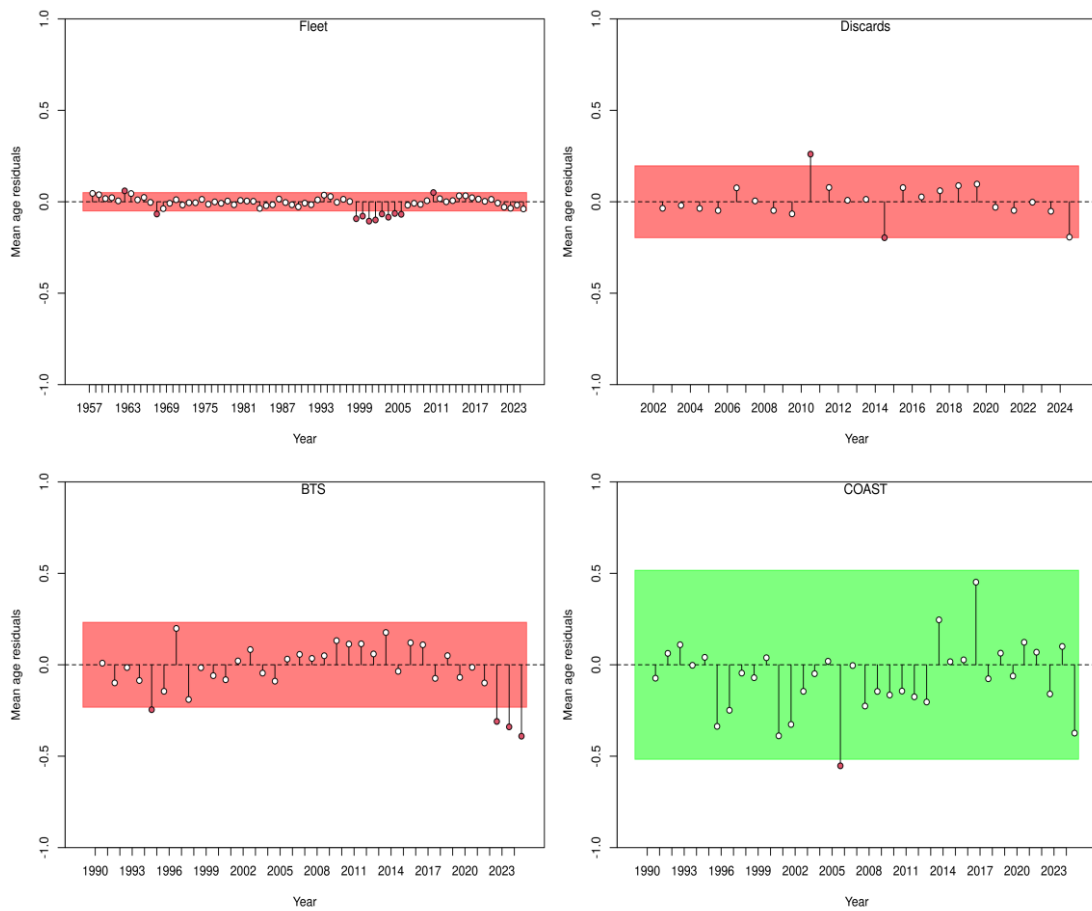


Figure 16.31. Sole in Subarea 4. Runs test of model fit to the mean age of the landings, discard and for the two indices of abundance: BTS and COAST. Green shading indicates no evidence and red shading evidence to reject the hypothesis of a randomly distributed time-series of residuals, respectively. The shaded area spans three residual standard deviations to either side of zero, and the red points outside of the shading violate the 'three-sigma limits' for that series.

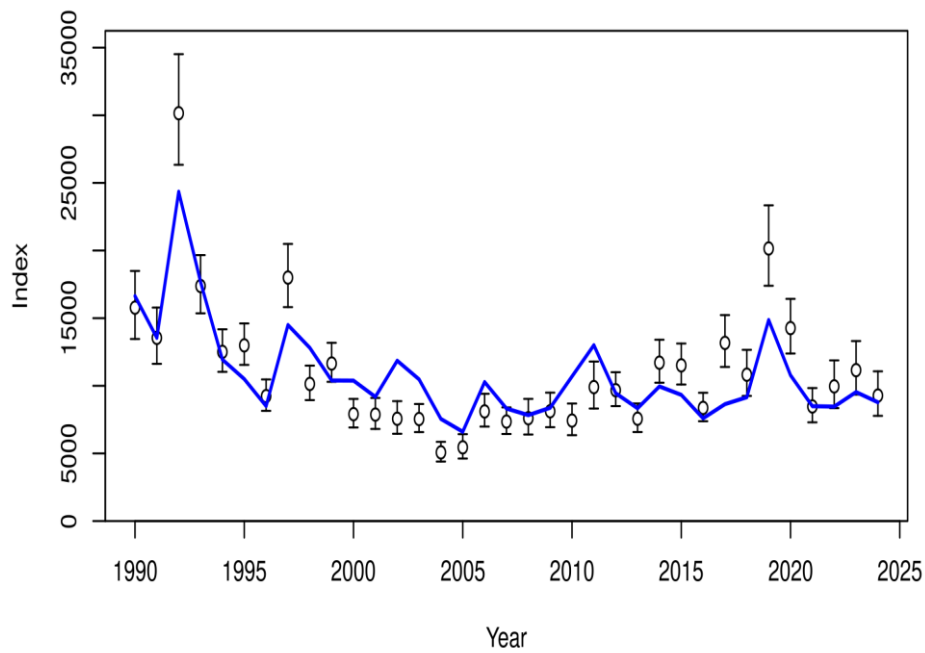


Figure 16.32. Sole in Subarea 4. Fit to the BTS index of abundance in biomass.

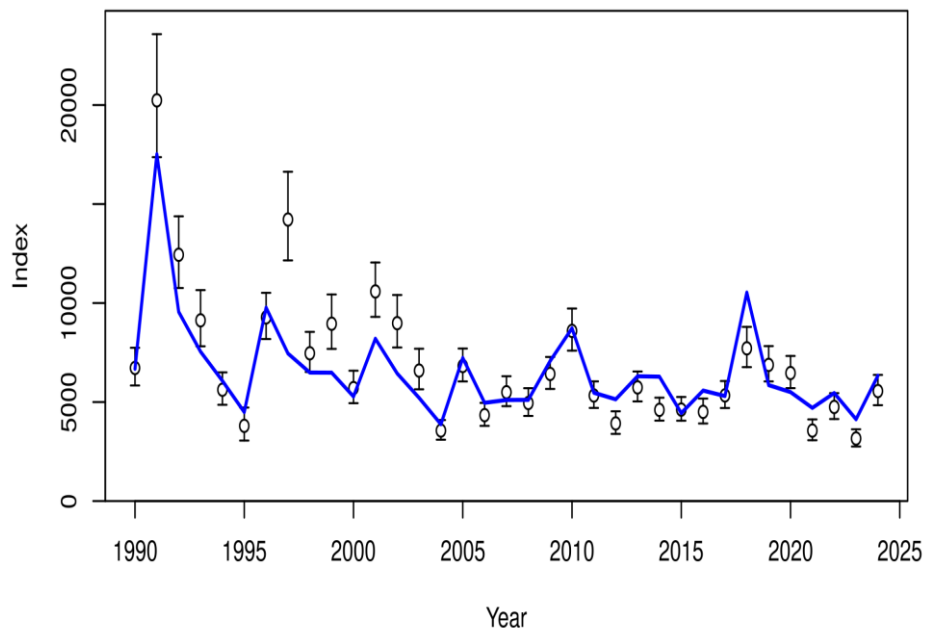


Figure 16.33. Sole in Subarea 4. Fit to the COAST (SNS + DYFS) index of abundance in biomass.

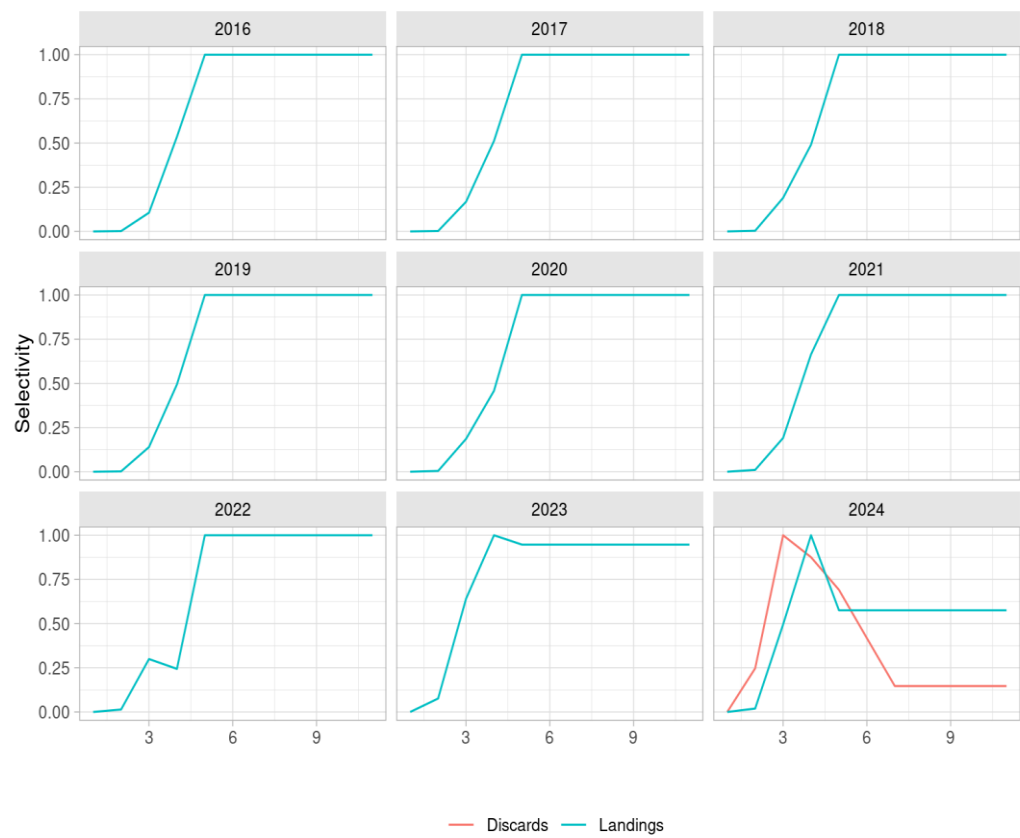


Figure 16.34. Sole in Subarea 4. Selectivity of the fleet at age for the period 2016–2024 as estimated by the SS3 model, with selectivity fixed to be constant from age 5 onwards. Landings selectivities are estimated annually, while discards have a single vector for the whole series

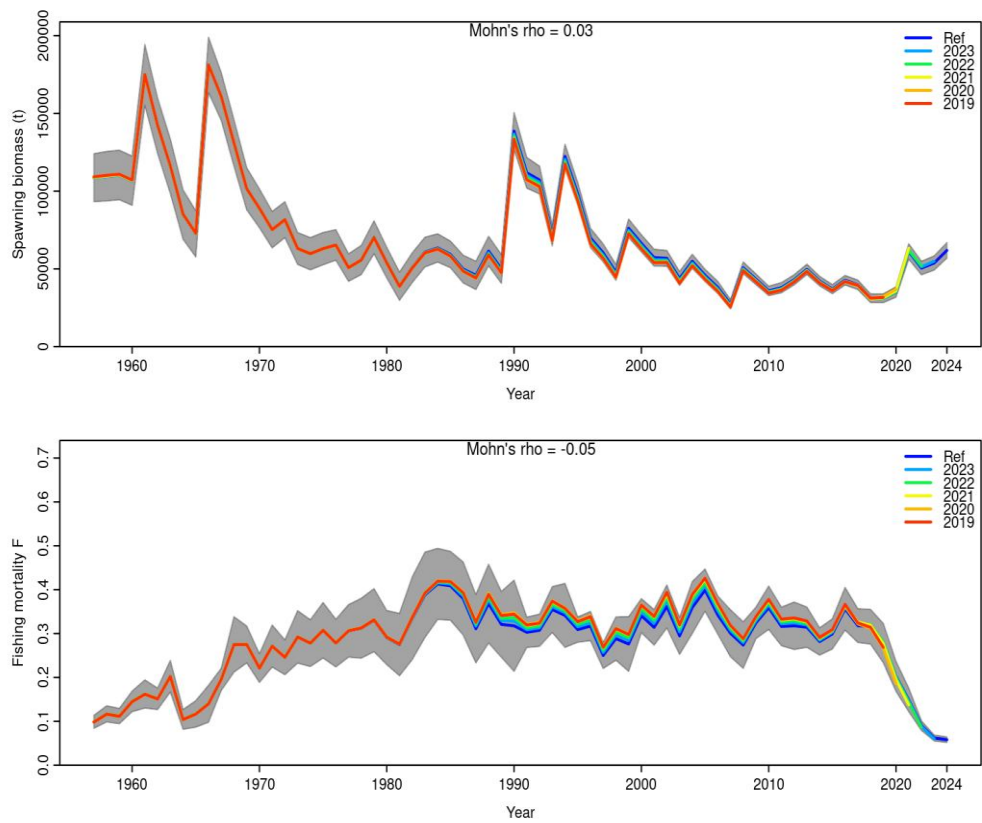


Figure 16.35. Sole in Subarea 4. Retrospective patterns in spawning stock biomass and mean fishing pressure, computed over five one-year steps.

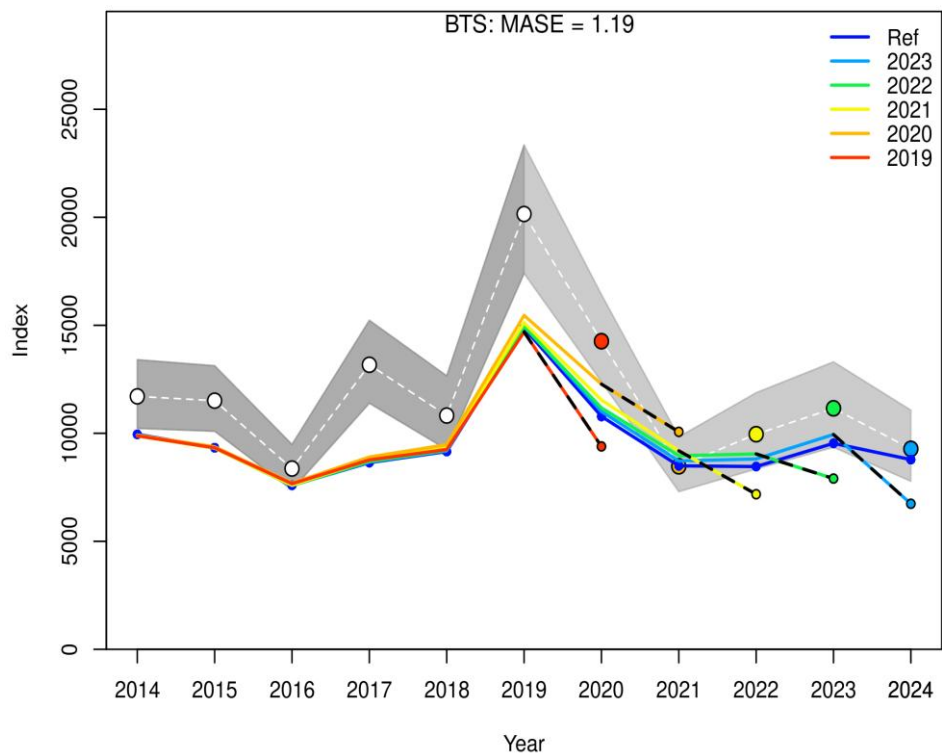


Figure 16.36. Sole in Subarea 4. Hindcasting cross-validation of the BTS index of abundance to estimate assessment model prediction skill.

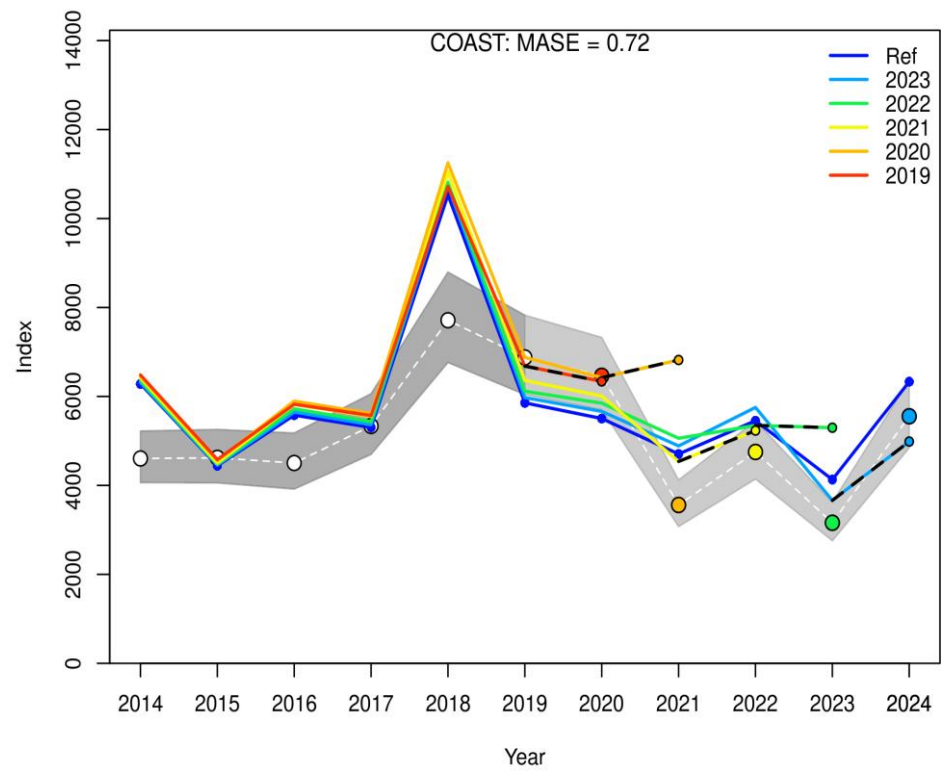


Figure 16.37. Sole in Subarea 4. Hindcasting cross-validation of the COAST index of abundance to estimate assessment model prediction skill.

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